

Bitwise linear trail through **LBlock** found by SAT

CiVerLy

July 20, 2026

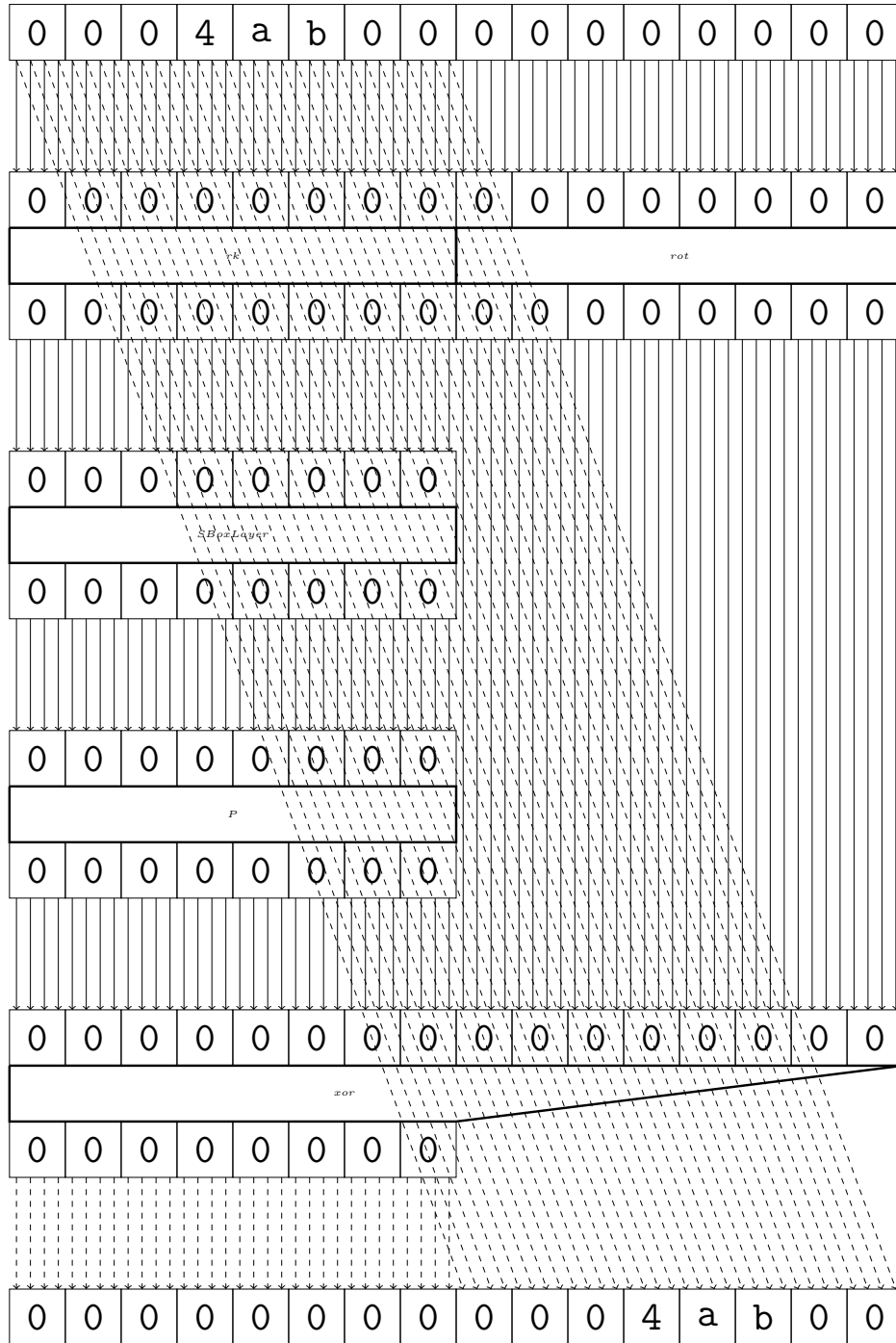
1 LBlock

Maximal linear correlation: 2^{-42}

0	0	0	0	4	a	b	0	0	0	0	0	0	0	0	0
0	0	0	4	a	b	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	4	a	b	0	0
0	0	0	0	0	0	0	0	0	0	0	4	a	b	0	0
0	4	a	b	0	0	0	0	8	0	c	1	0	0	0	0
0	4	a	b	0	0	0	0	8	0	c	1	0	0	0	0
c	1	0	0	0	8	0	0	a	b	0	4	0	0	0	0
c	1	0	0	0	8	0	0	a	b	0	4	0	0	0	0
a	0	4	0	0	0	0	a	b	0	0	0	0	0	0	0
a	0	4	0	0	0	0	a	b	0	0	0	0	0	0	0
0	0	0	0	0	0	0	a	0	4	0	0	0	a		
0	0	0	0	0	0	0	a	0	4	0	0	0	a		
4	0	0	0	a	b	0	0	c	0	0	4	0	0	5	
4	0	0	0	a	b	0	0	c	0	0	4	0	0	5	
0	0	4	0	0	5	0	c	0	0	0	0	a	=	f	
0	0	4	0	0	5	0	c	0	0	0	0	a	=	f	
0	0	0	a	=	f	0	0	0	0	0	0	0	0	0	0
0	0	0	a	=	f	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	a	=	f	0	0	
0	0	0	0	0	0	0	0	0	0	a	=	f	0	0	
0	a	=	f	0	0	0	0	c	0	c	0	0	0	0	
0	a	=	f	0	0	0	0	c	0	c	0	0	0	0	
c	c	0	0	0	0	c	0	0	a	=	0	4	0	0	0
c	c	0	0	0	0	c	0	0	a	=	0	4	0	0	0
e	0	4	0	0	0	a	0	0	0	0	0	0	0	0	0
e	0	4	0	0	0	a	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	a	0	4	0	0	0	a		
0	0	0	0	0	0	0	e	0	4	0	0	0	e		
4	0	0	0	a	=	0	0	c	0	0	4	0	0	4	
4	0	0	0	a	=	0	0	c	0	0	4	0	0	4	
0	0	4	0	0	4	0	c	0	0	0	0	a	=	0	4
0	0	4	0	0	4	0	c	0	0	0	0	a	=	0	4
0	0	0	a	=	0	4	0	0	0	0	0	0	4	0	0
0	0	0	a	=	0	4	0	0	0	0	0	0	0	4	0
0	0	0	a	=	0	4	0	0	0	0	0	0	0	0	4
0	0	0	4	0	0	0	0	0	0	9	a	0	4	0	0
0	0	0	4	0	0	0	0	0	0	9	a	0	4	0	0
9	a	0	4	0	0	0	0	0	5	d	0	0	0	0	0
9	a	0	4	0	0	0	0	0	5	d	0	0	0	0	0
d	0	0	0	0	0	0	5	9	a	0	4	0	0	4	0
d	0	0	0	0	0	0	5	9	a	0	4	0	0	4	0
9	a	0	4	0	0	4	0	d	0	0	0	0	0	0	5
9	a	0	4	0	0	4	0	d	0	0	0	0	0	0	5

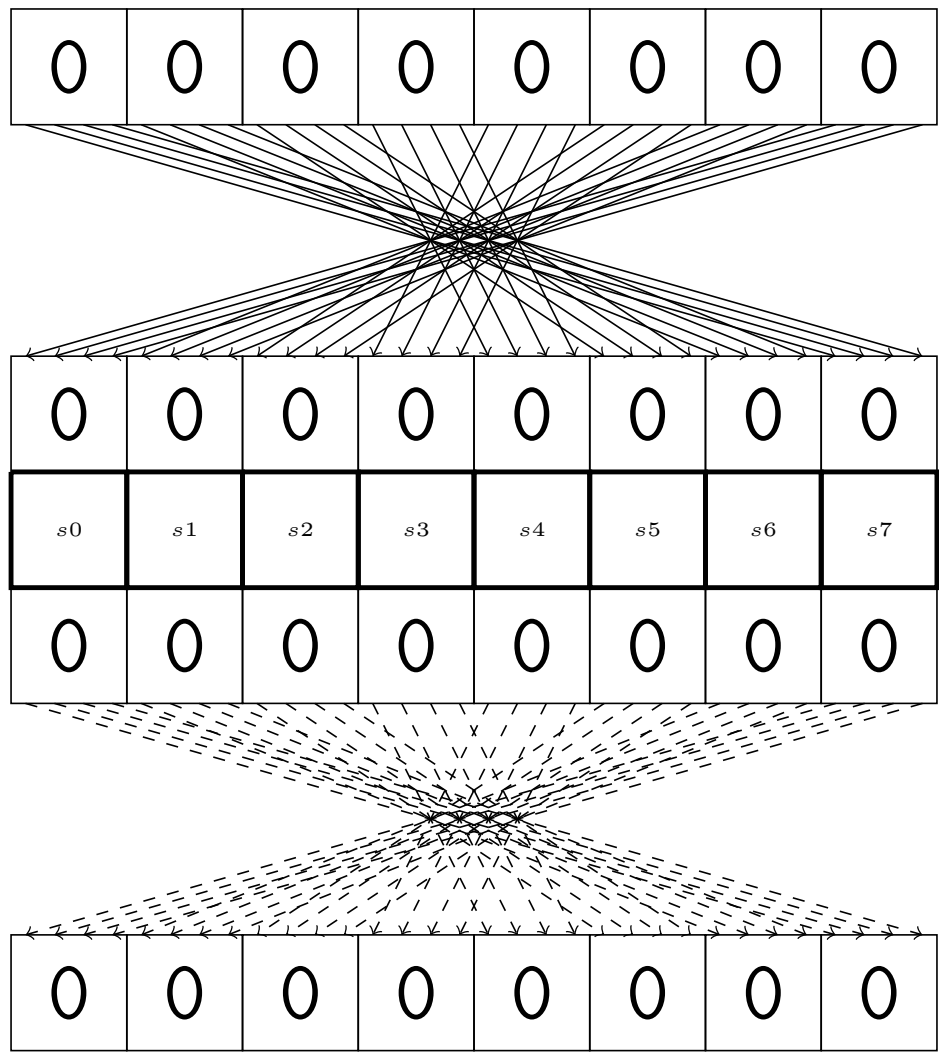
2 LBlock_round

Maximal linear correlation: $2^{-0.0}$



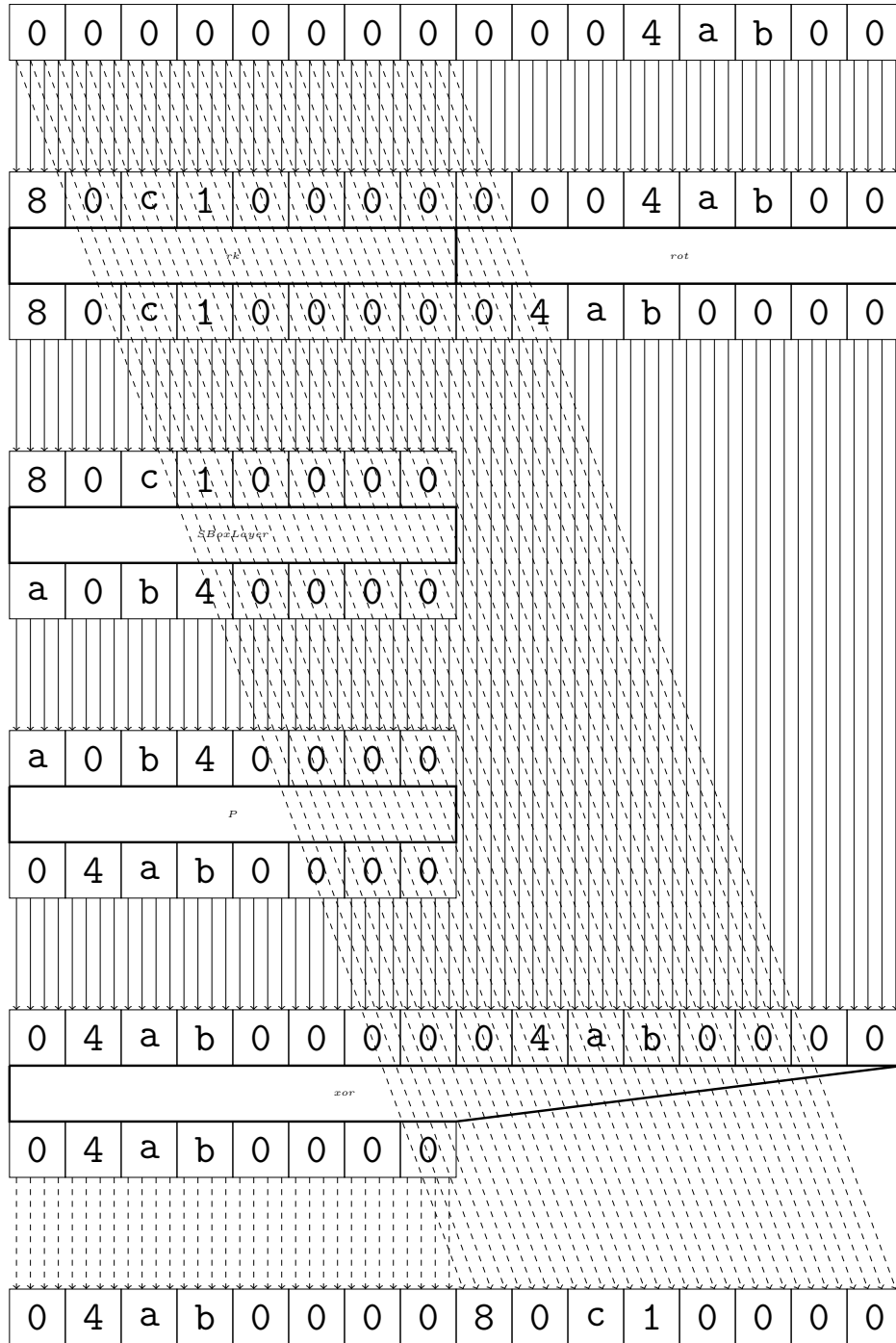
3 SBoxLayer

Maximal linear correlation: $2^{-0.0}$



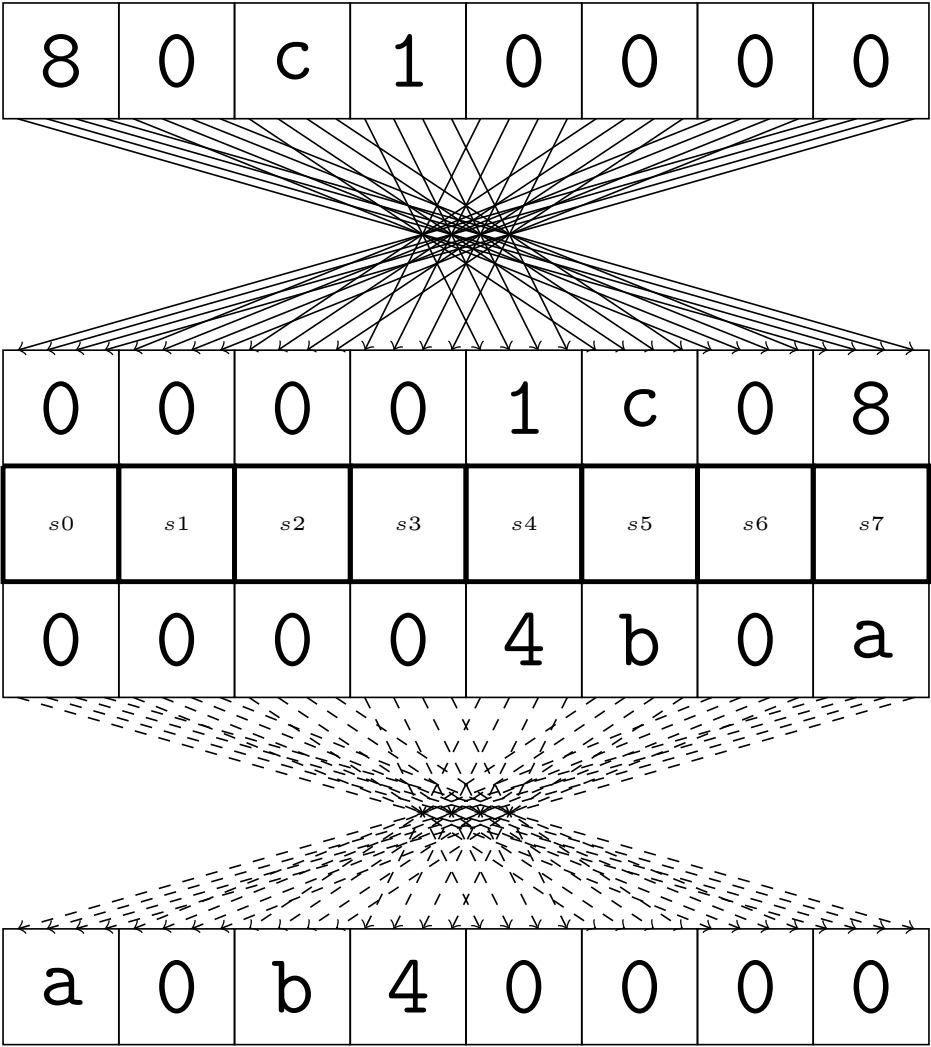
4 LBlock_round

Maximal linear correlation: $2^{-3.0}$



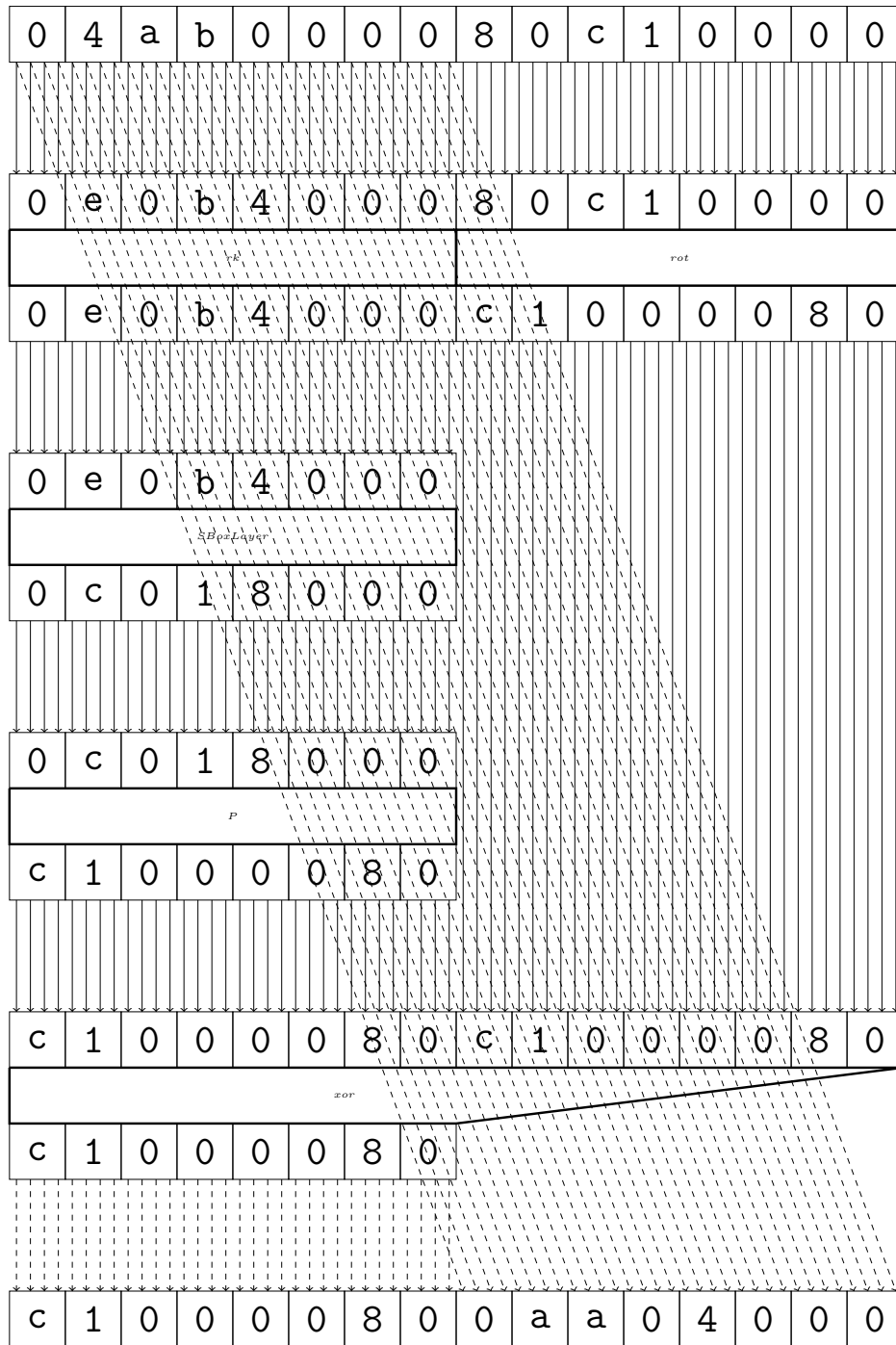
5 SBoxLayer

Maximal linear correlation: $2^{-3.0}$



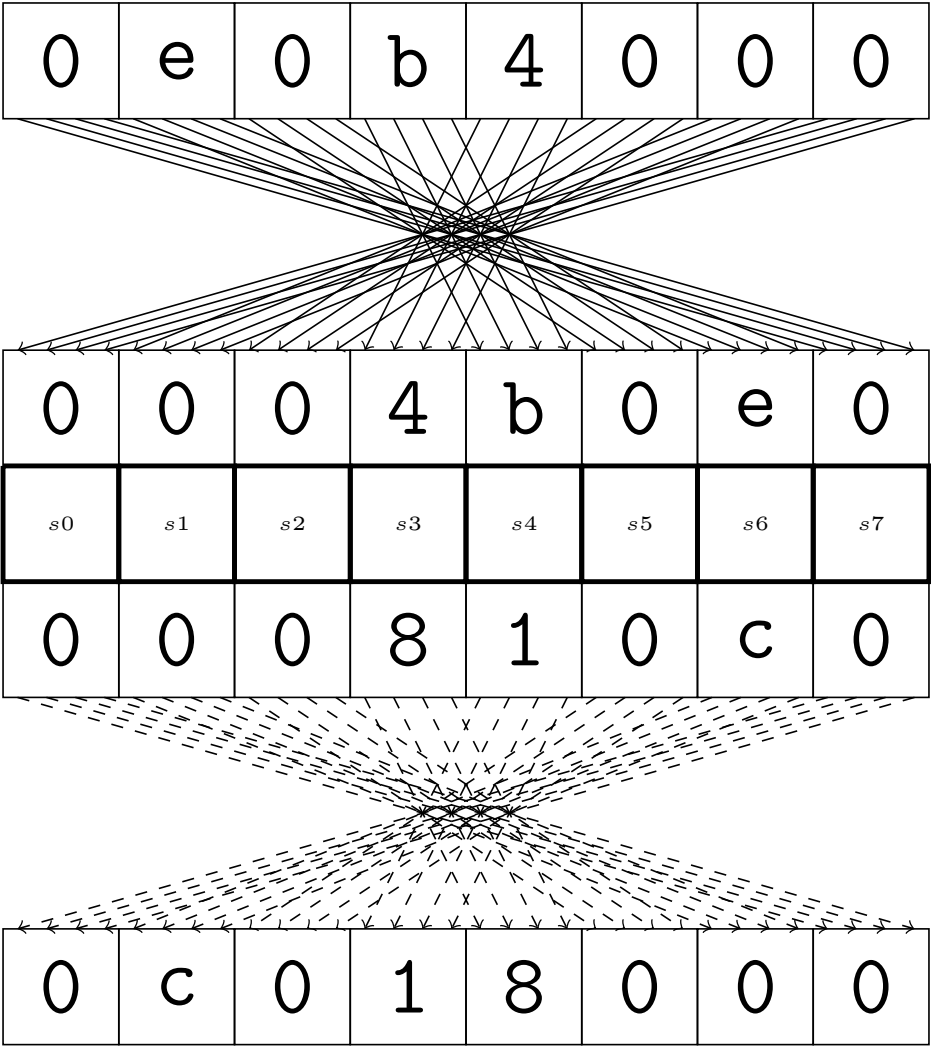
6 LBlock_round

Maximal linear correlation: $2^{-3.0}$



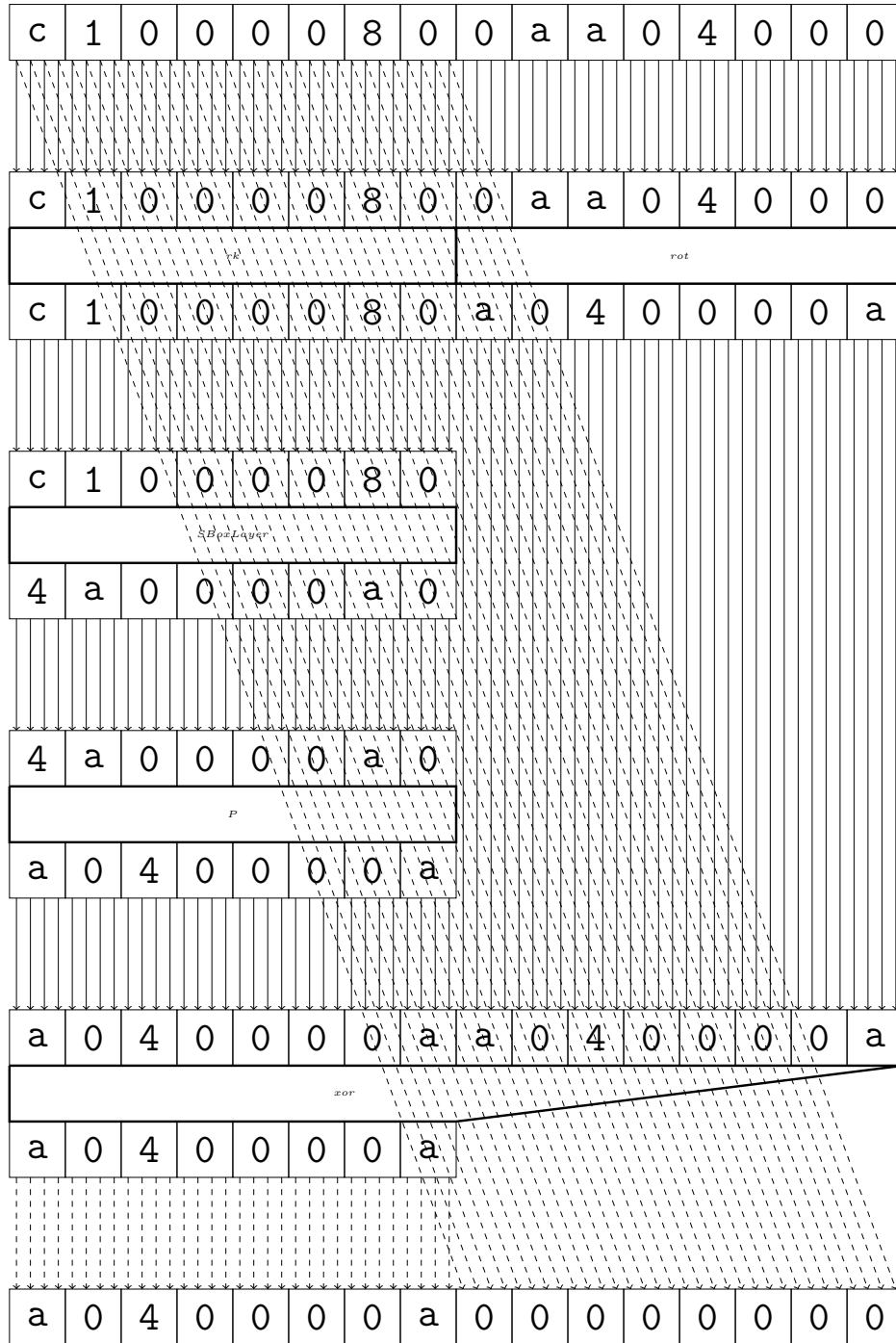
7 SBoxLayer

Maximal linear correlation: $2^{-3.0}$



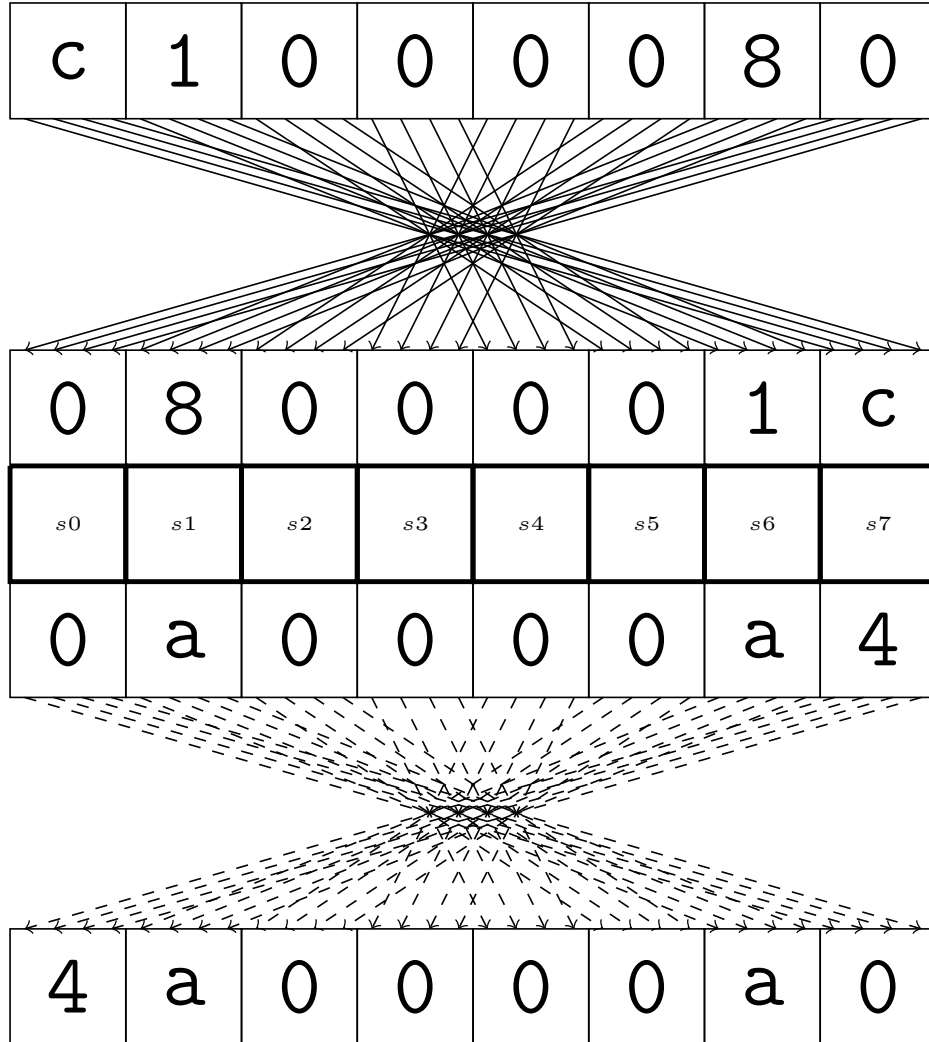
8 LBlock_round

Maximal linear correlation: $2^{-3.0}$



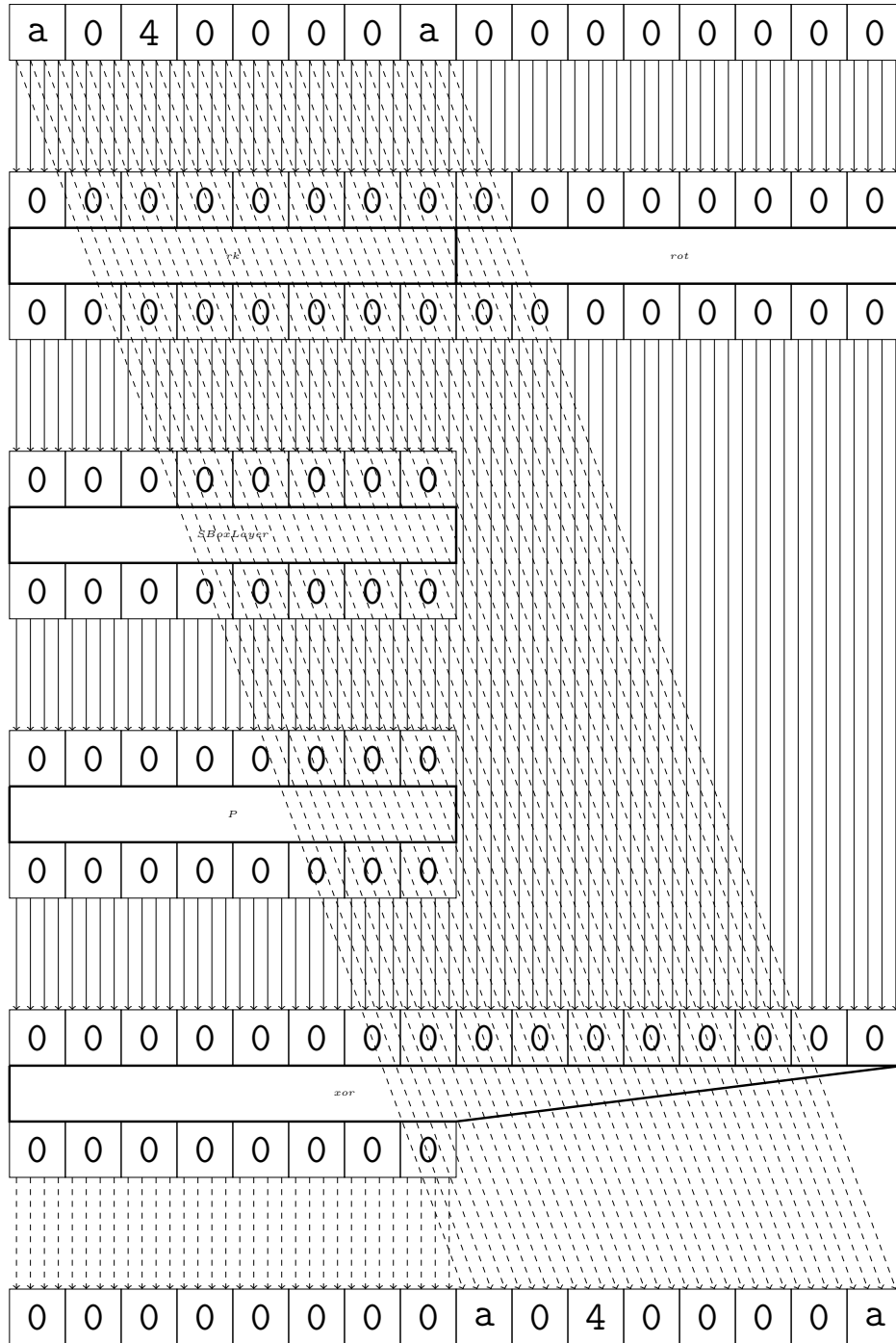
9 SBoxLayer

Maximal linear correlation: $2^{-3.0}$



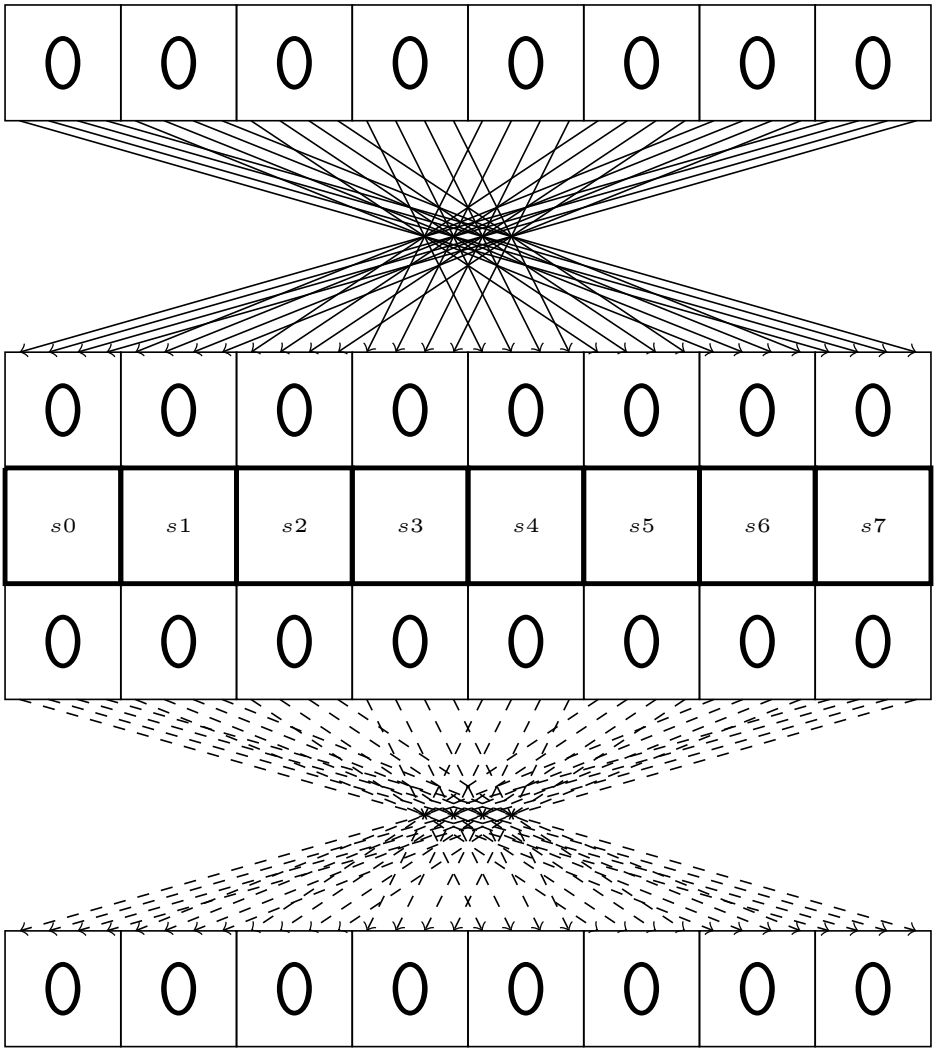
10 LBlock_round

Maximal linear correlation: $2^{-0.0}$



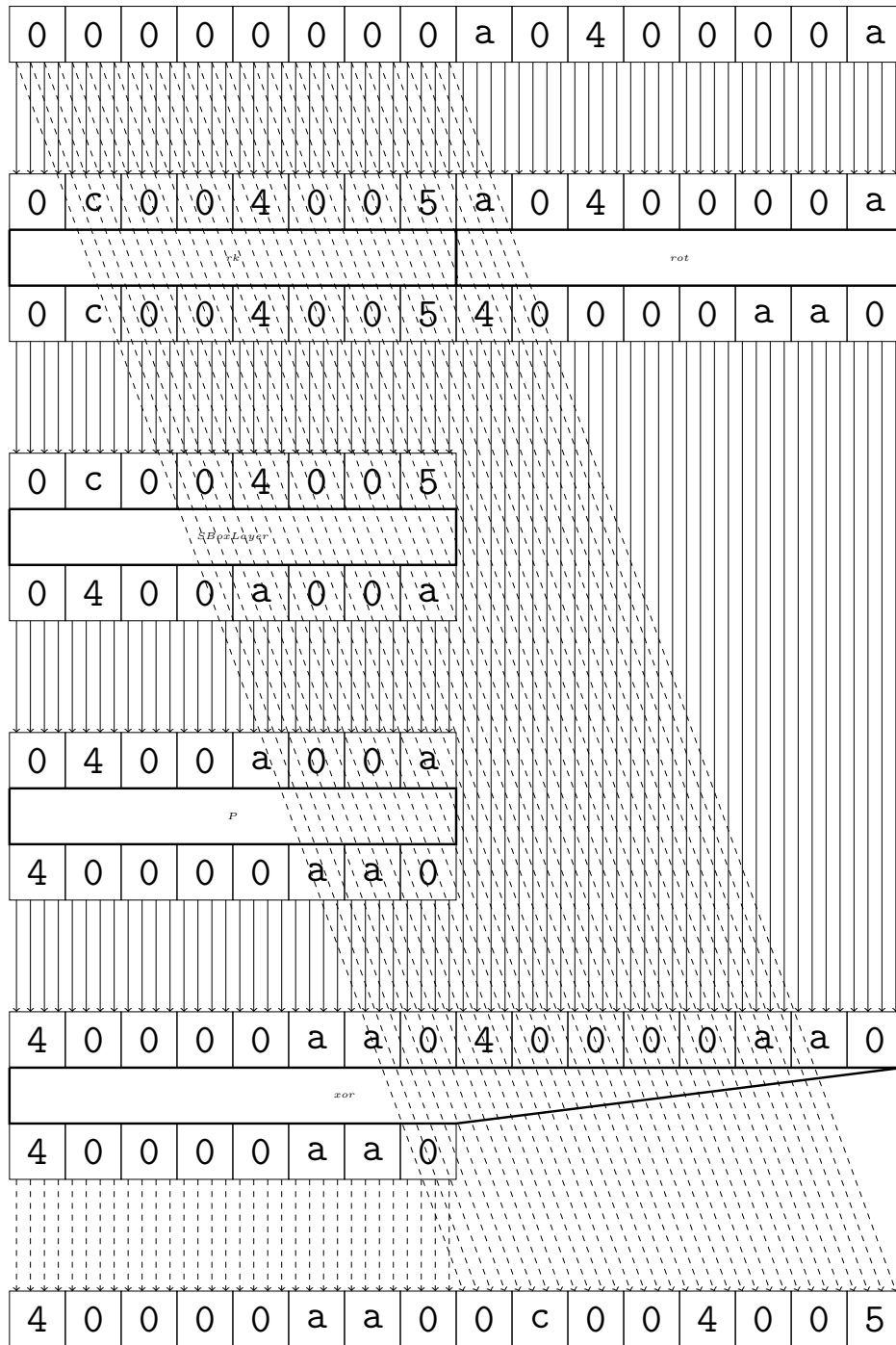
11 SBoxLayer

Maximal linear correlation: $2^{-0.0}$



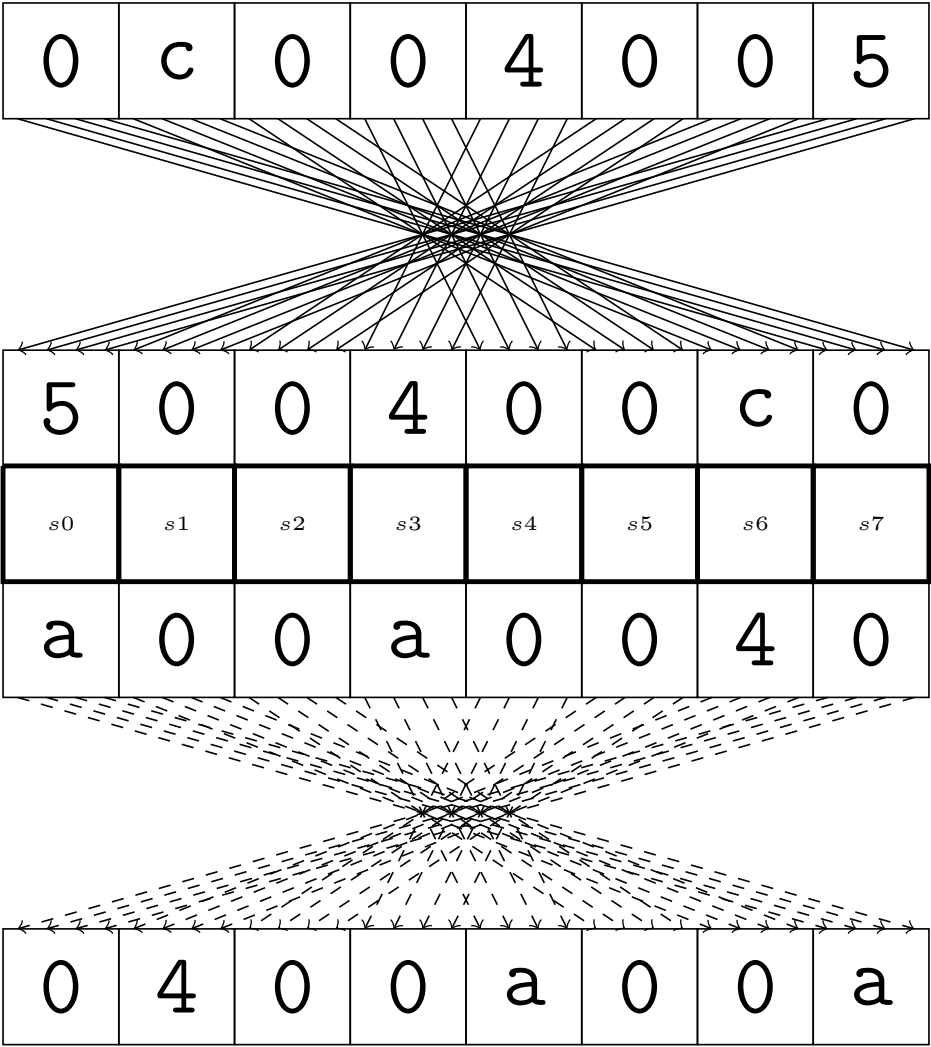
12 LBlock_round

Maximal linear correlation: $2^{-3.0}$

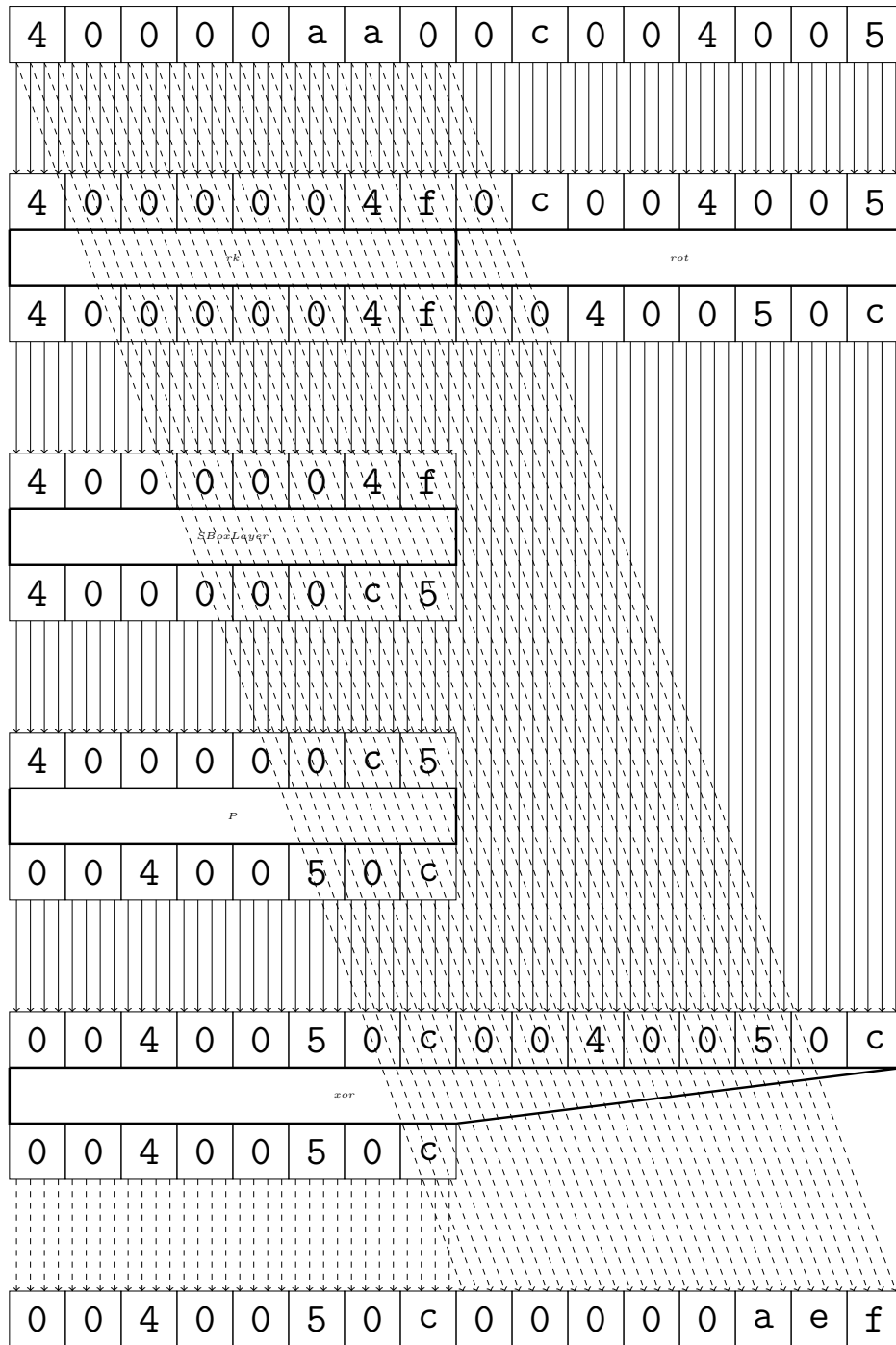


13 SBoxLayer

Maximal linear correlation: $2^{-3.0}$

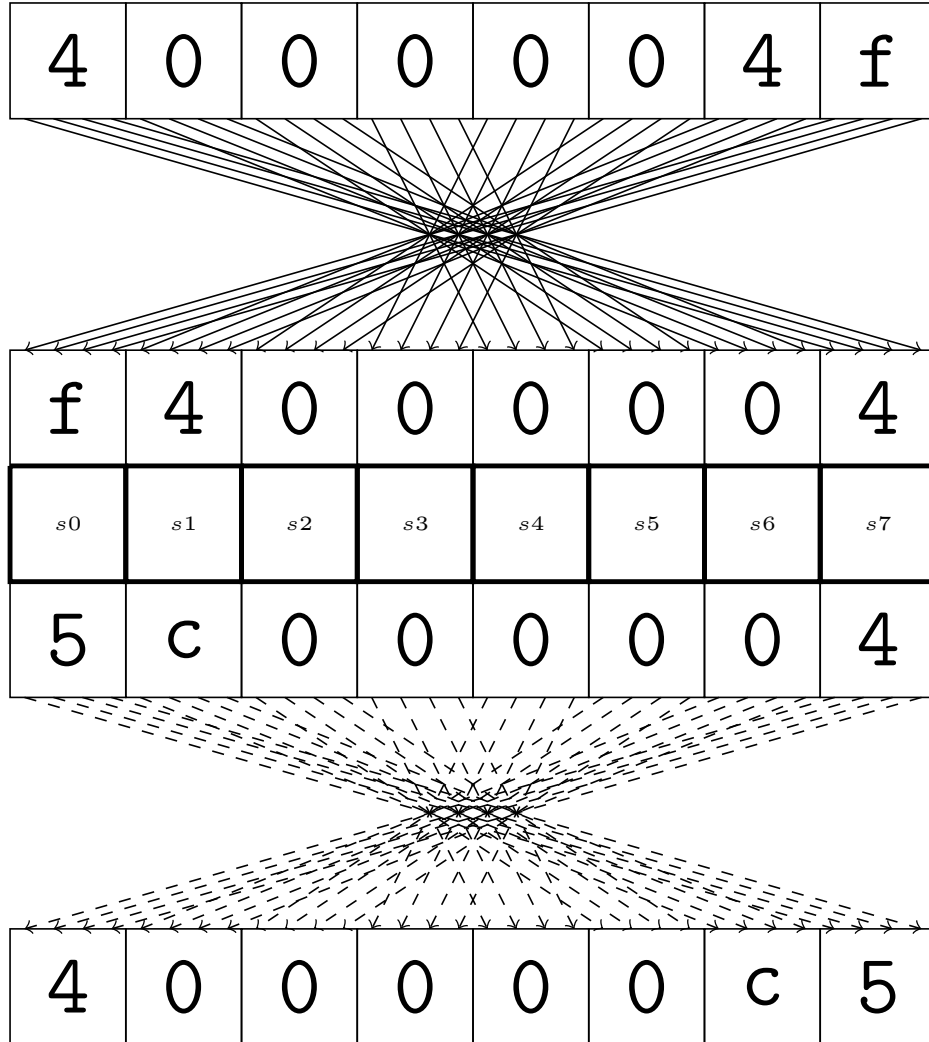


14 LBlock_round

Maximal linear correlation: $2^{-3.0}$ 

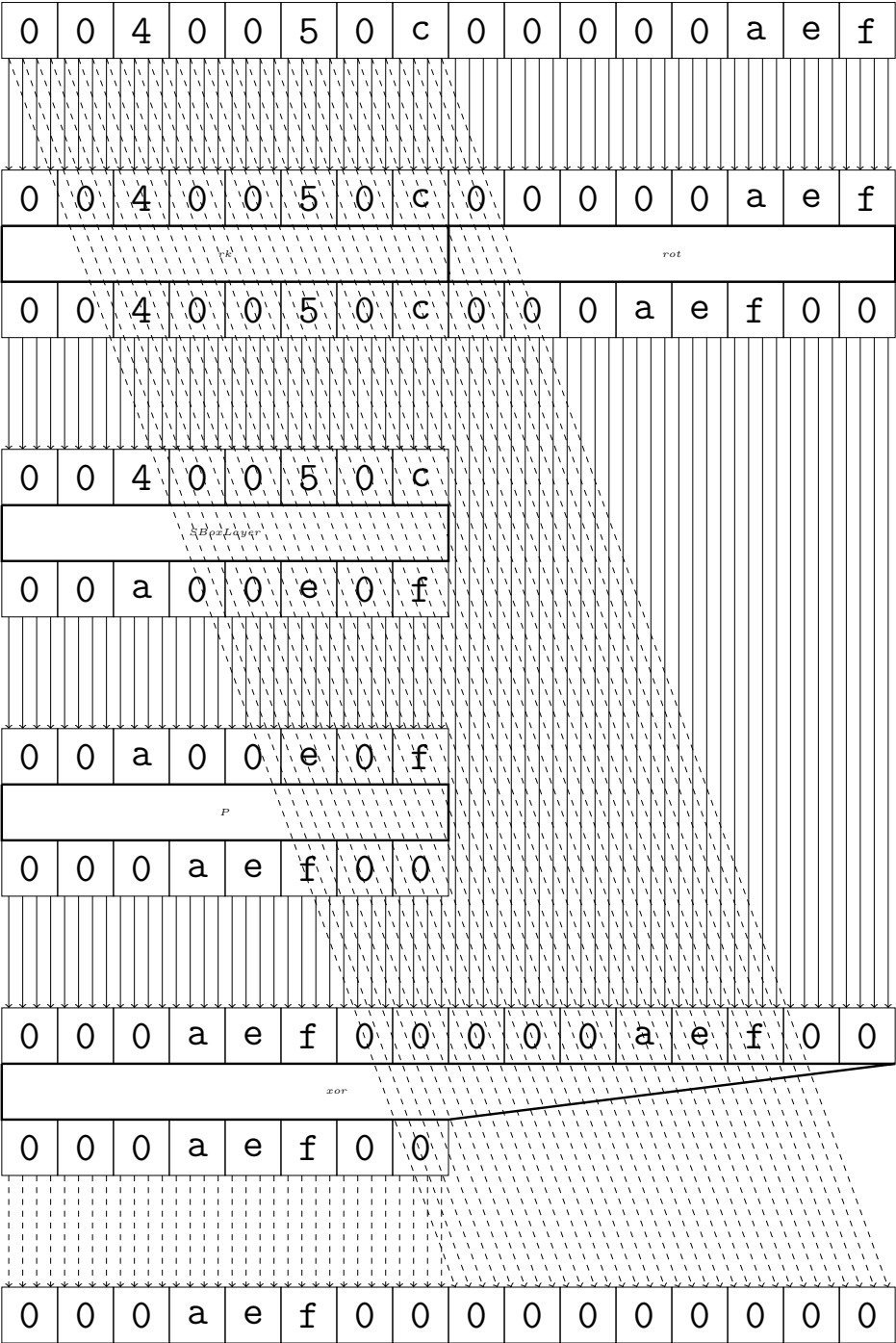
15 SBoxLayer

Maximal linear correlation: $2^{-3.0}$



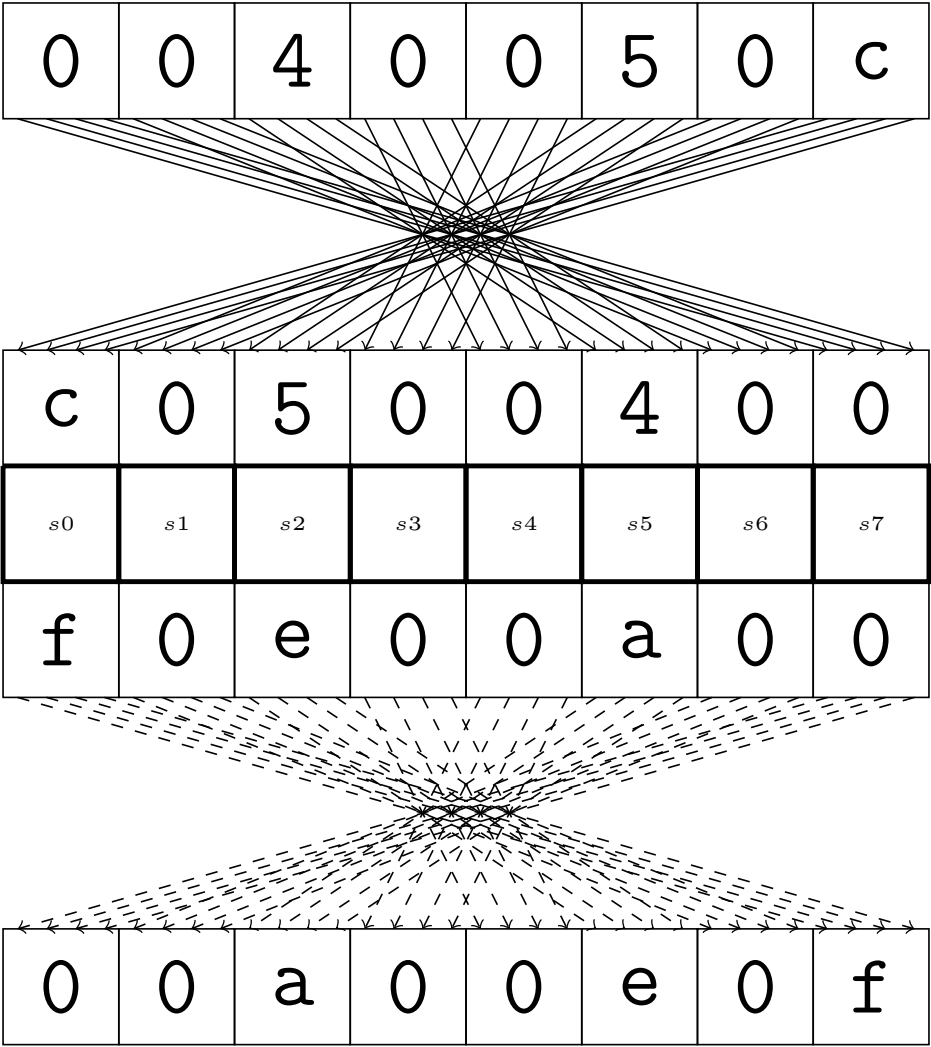
16 LBlock_round

Maximal linear correlation: $2^{-3.0}$

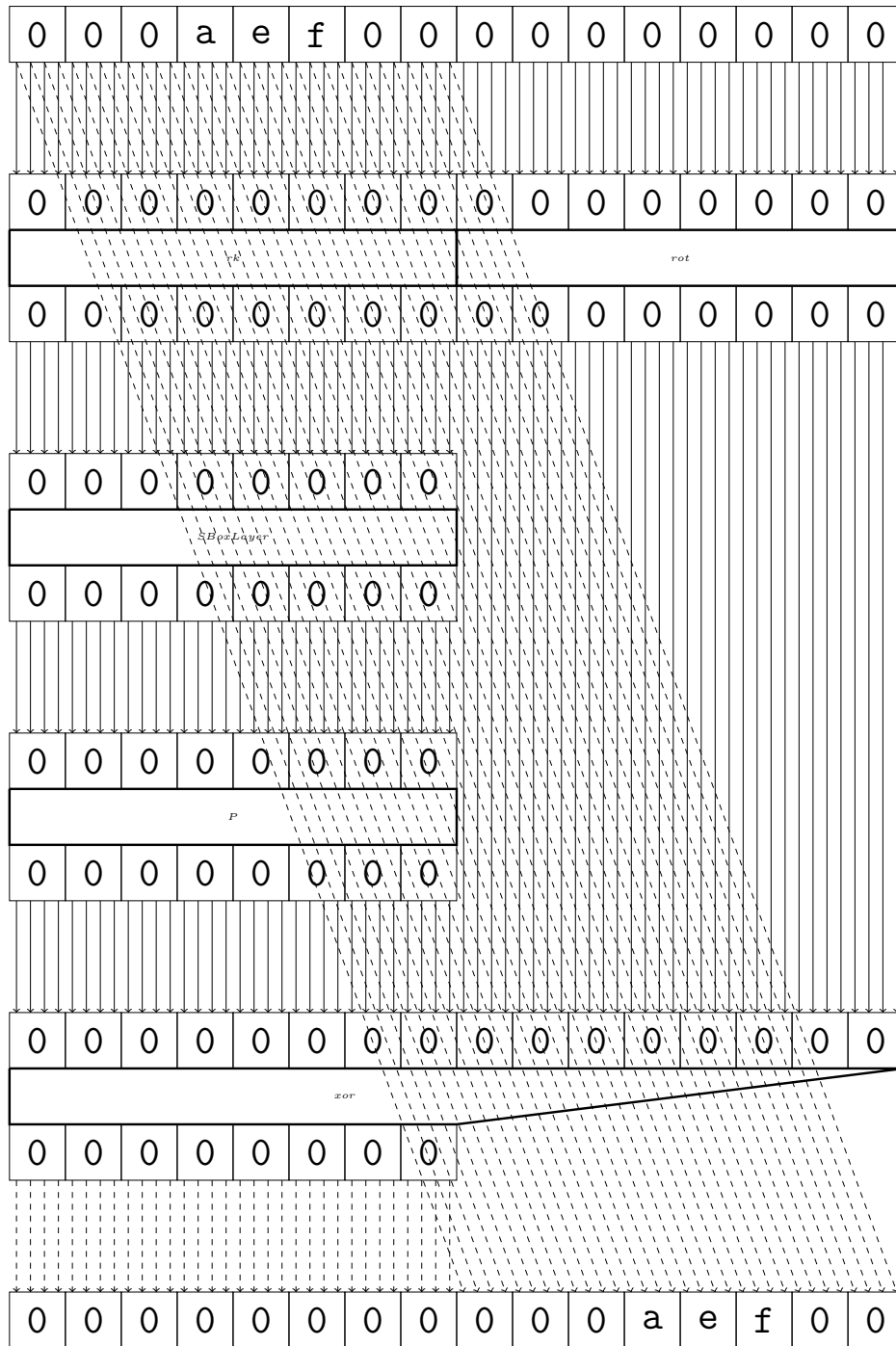


17 SBoxLayer

Maximal linear correlation: $2^{-3.0}$

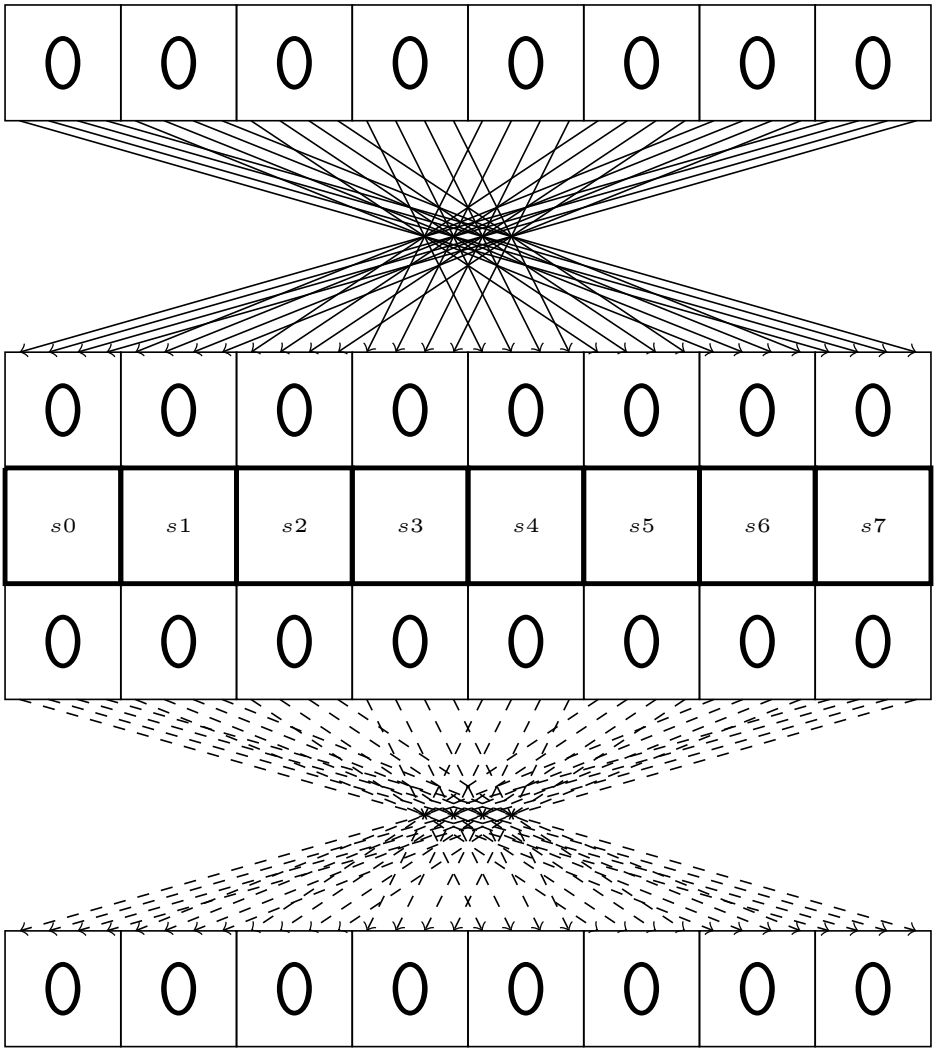


18 LBlock_round

Maximal linear correlation: $2^{-0.0}$ 

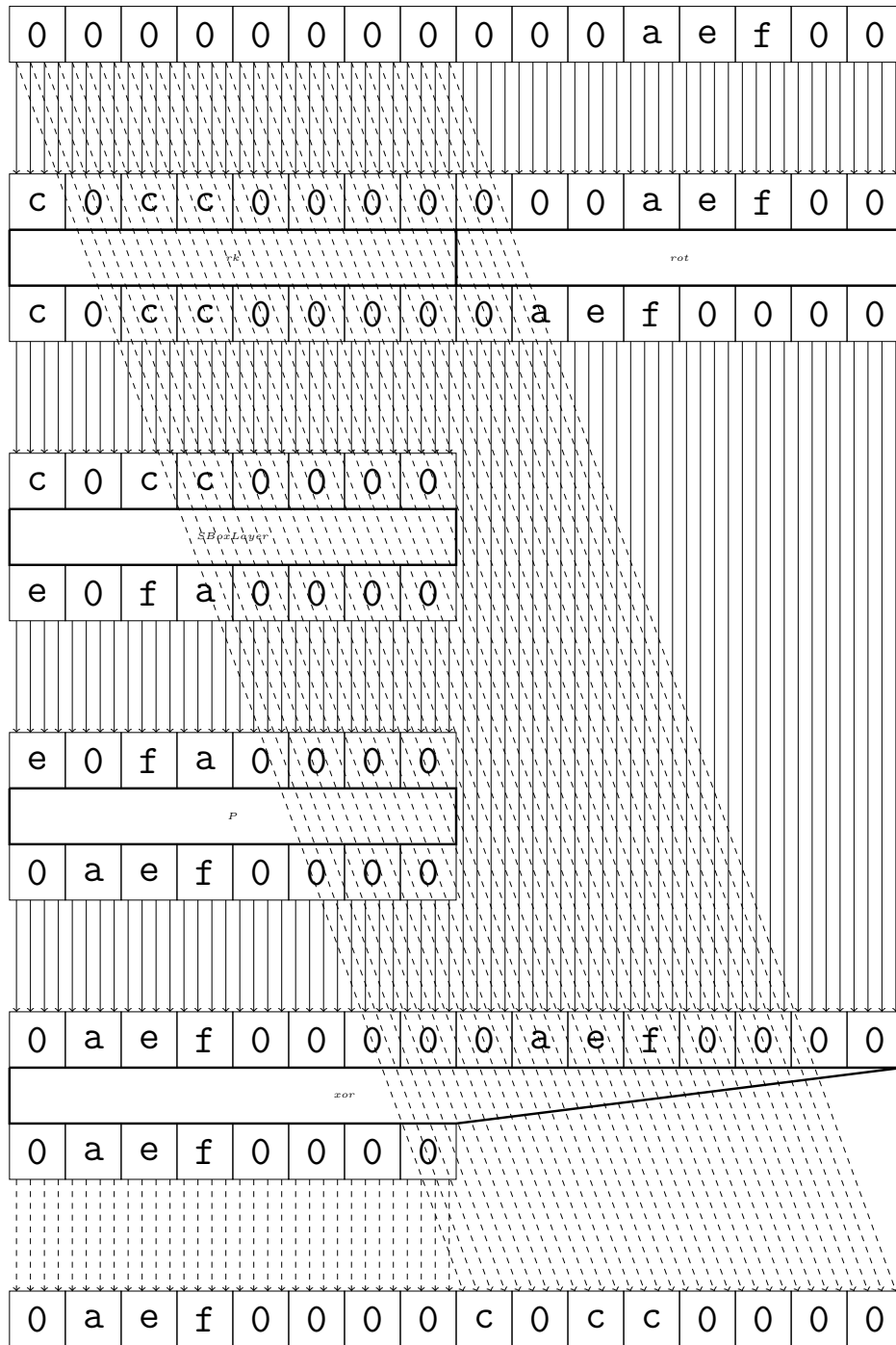
19 SBoxLayer

Maximal linear correlation: $2^{-0.0}$



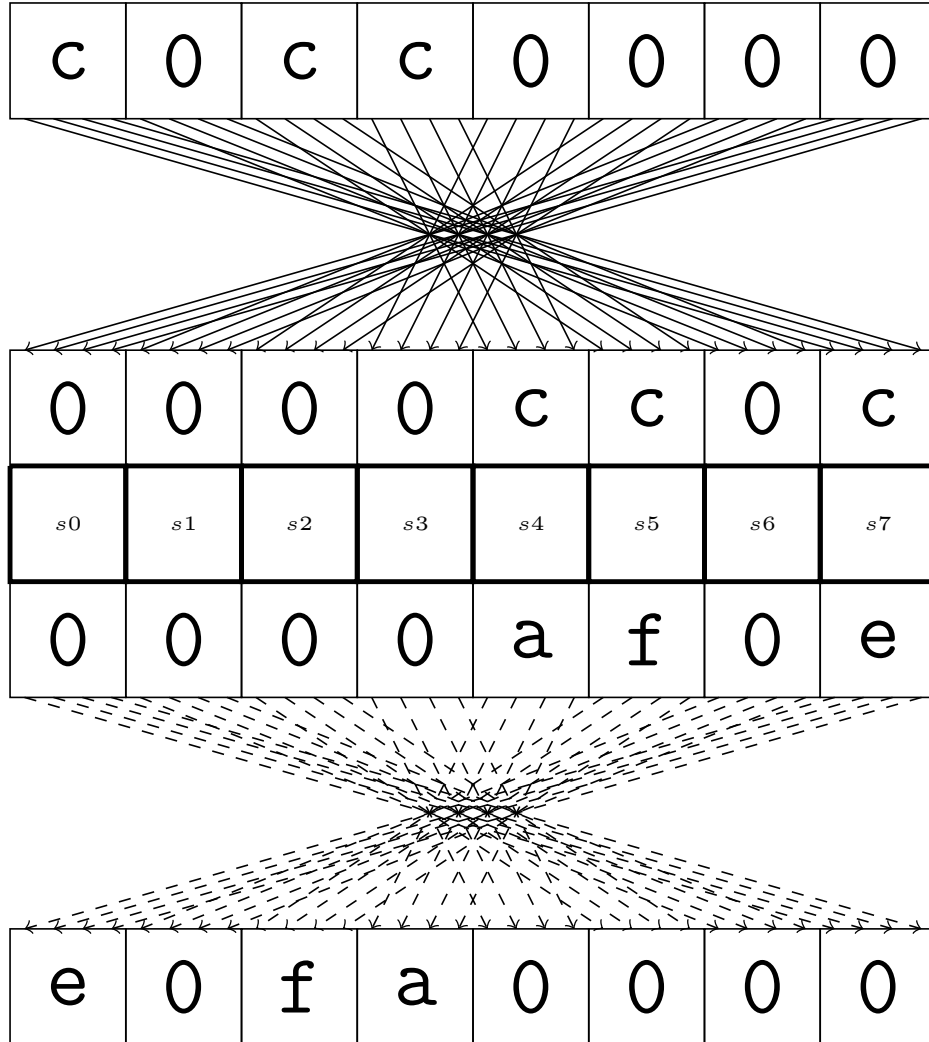
20 LBlock_round

Maximal linear correlation: $2^{-3.0}$



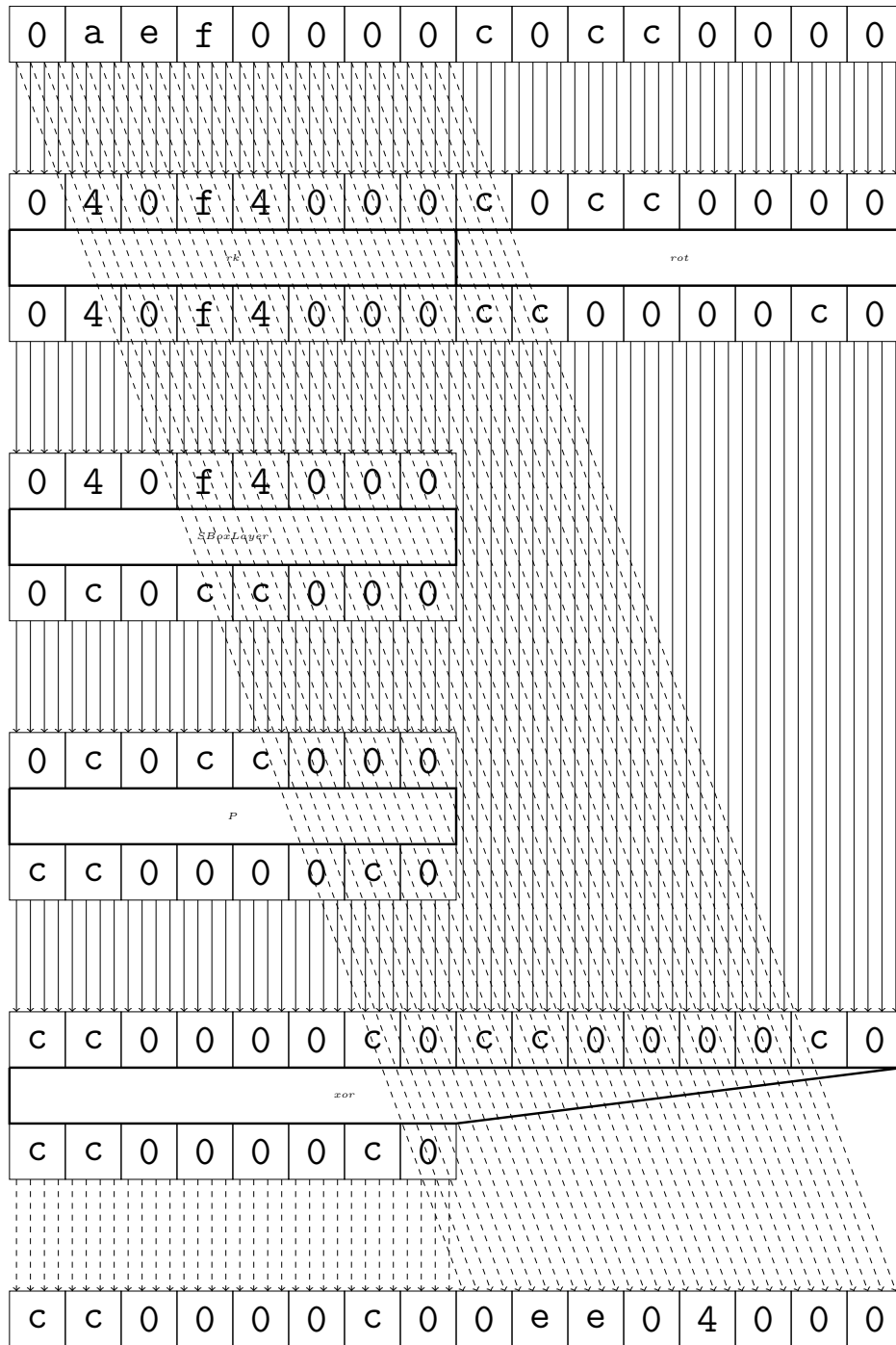
21 SBoxLayer

Maximal linear correlation: $2^{-3.0}$



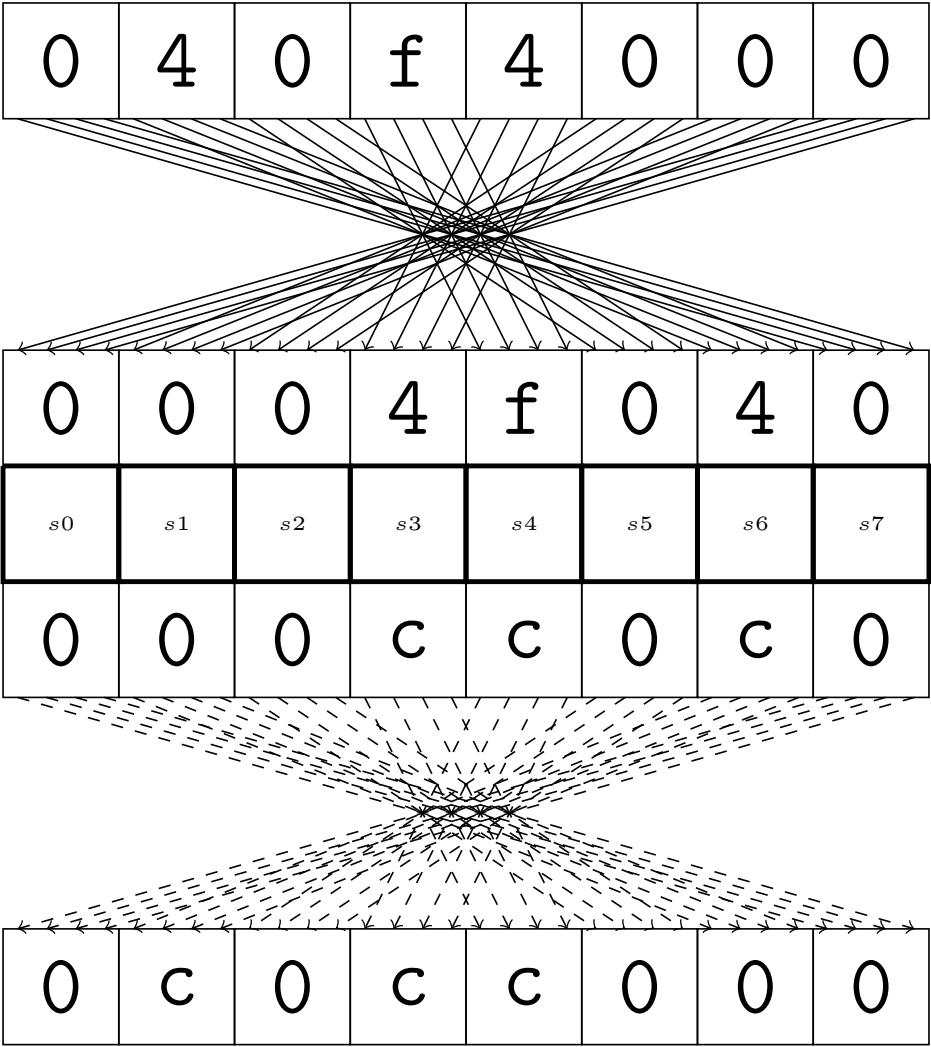
22 LBlock_round

Maximal linear correlation: $2^{-4.0}$



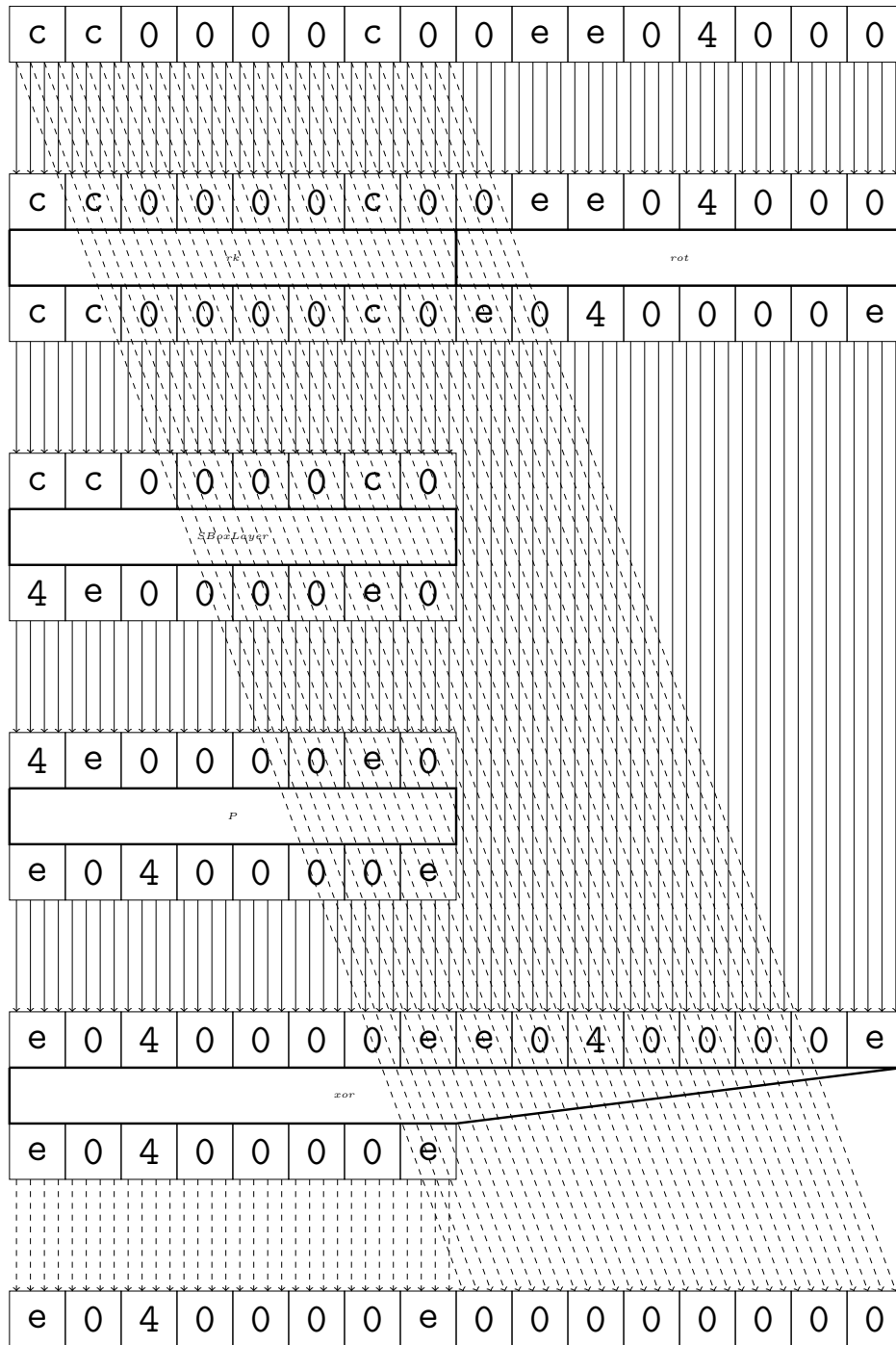
23 SBoxLayer

Maximal linear correlation: $2^{-4.0}$



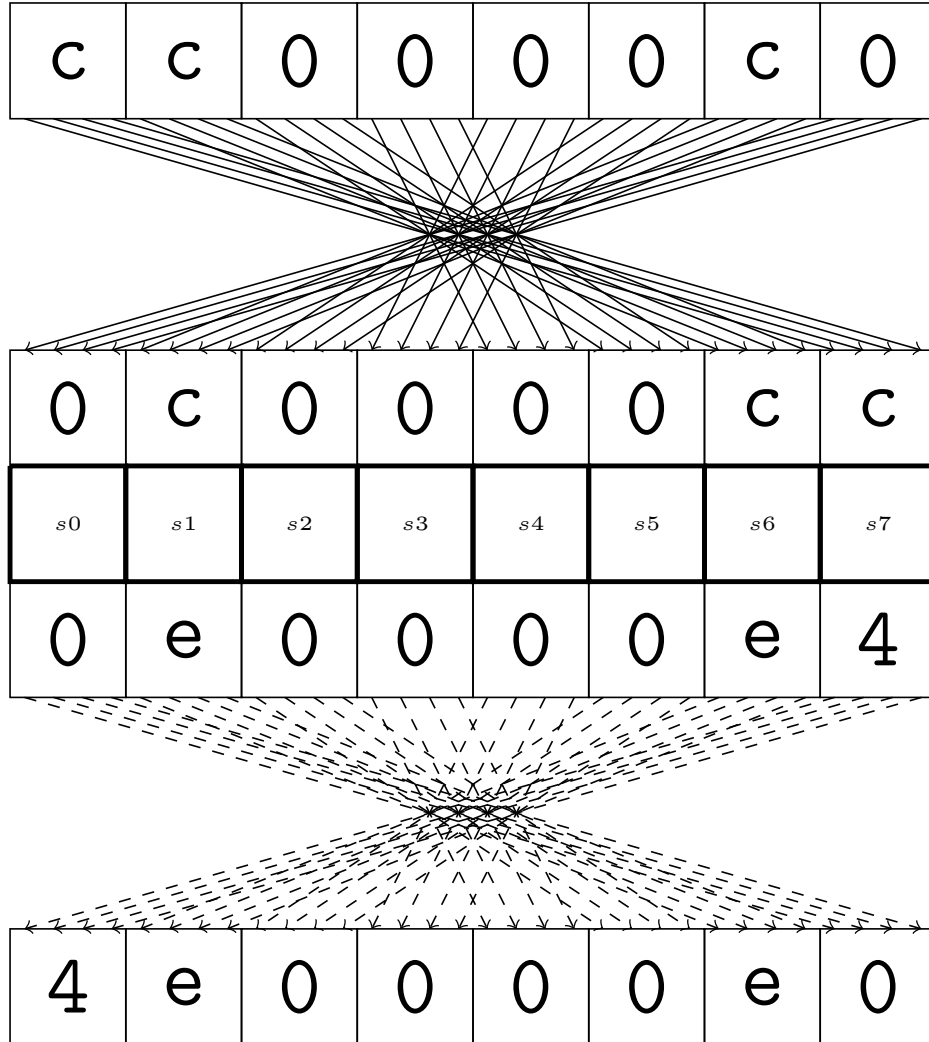
24 LBlock_round

Maximal linear correlation: $2^{-3.0}$



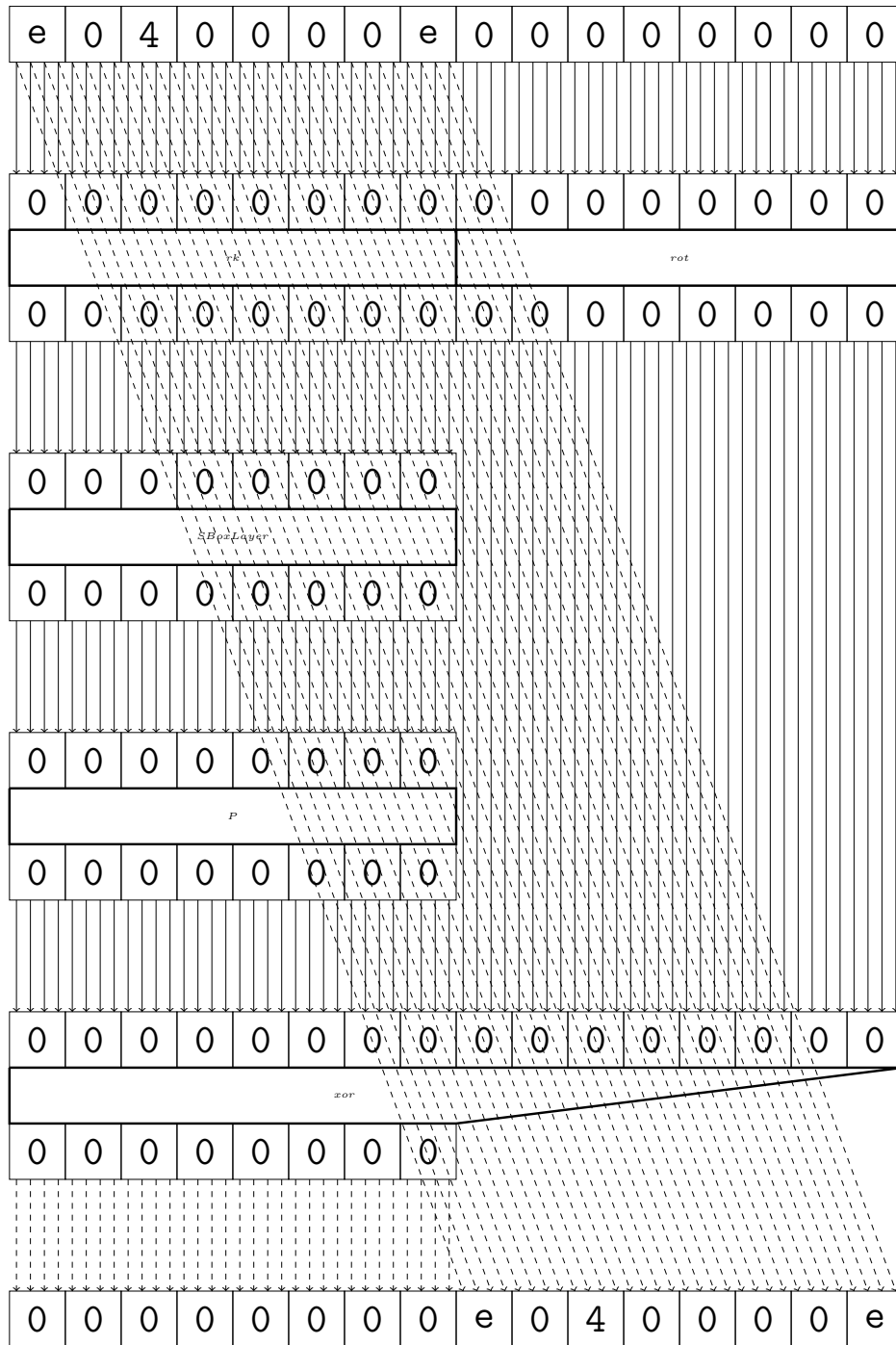
25 SBoxLayer

Maximal linear correlation: $2^{-3.0}$



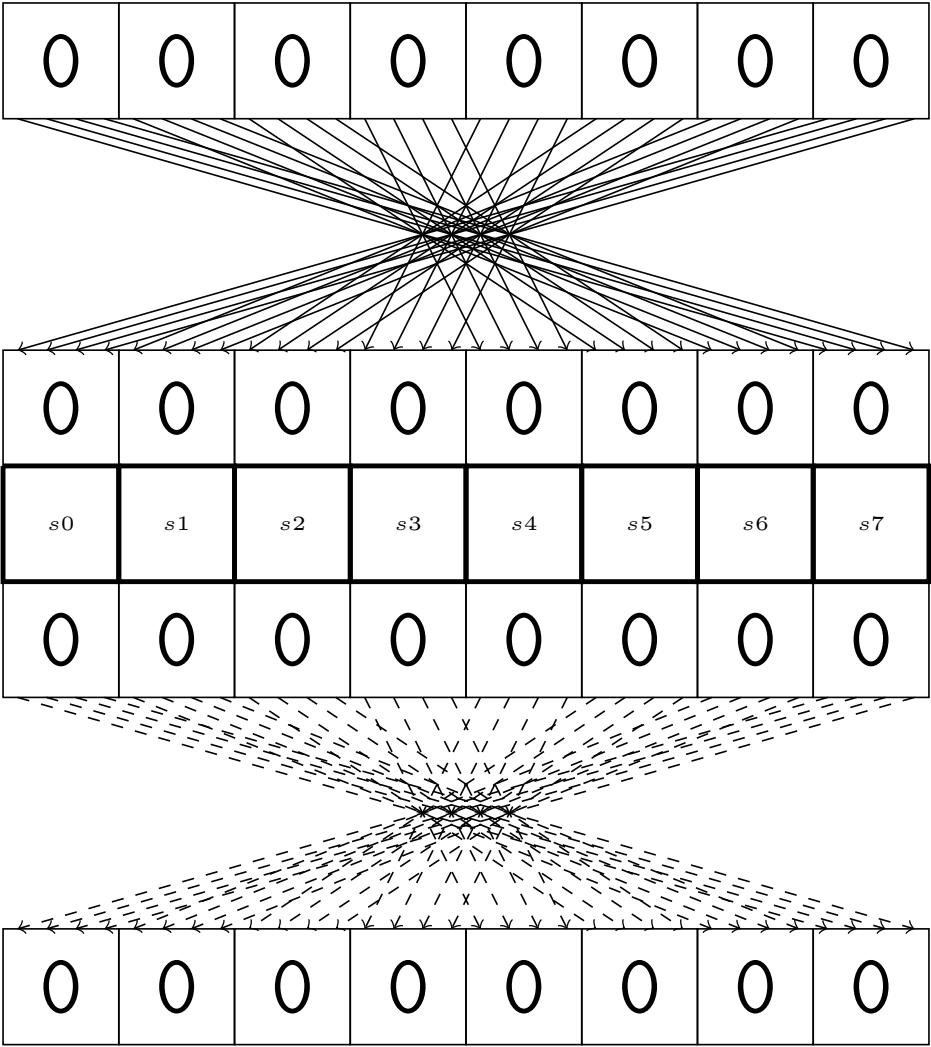
26 LBlock_round

Maximal linear correlation: $2^{-0.0}$



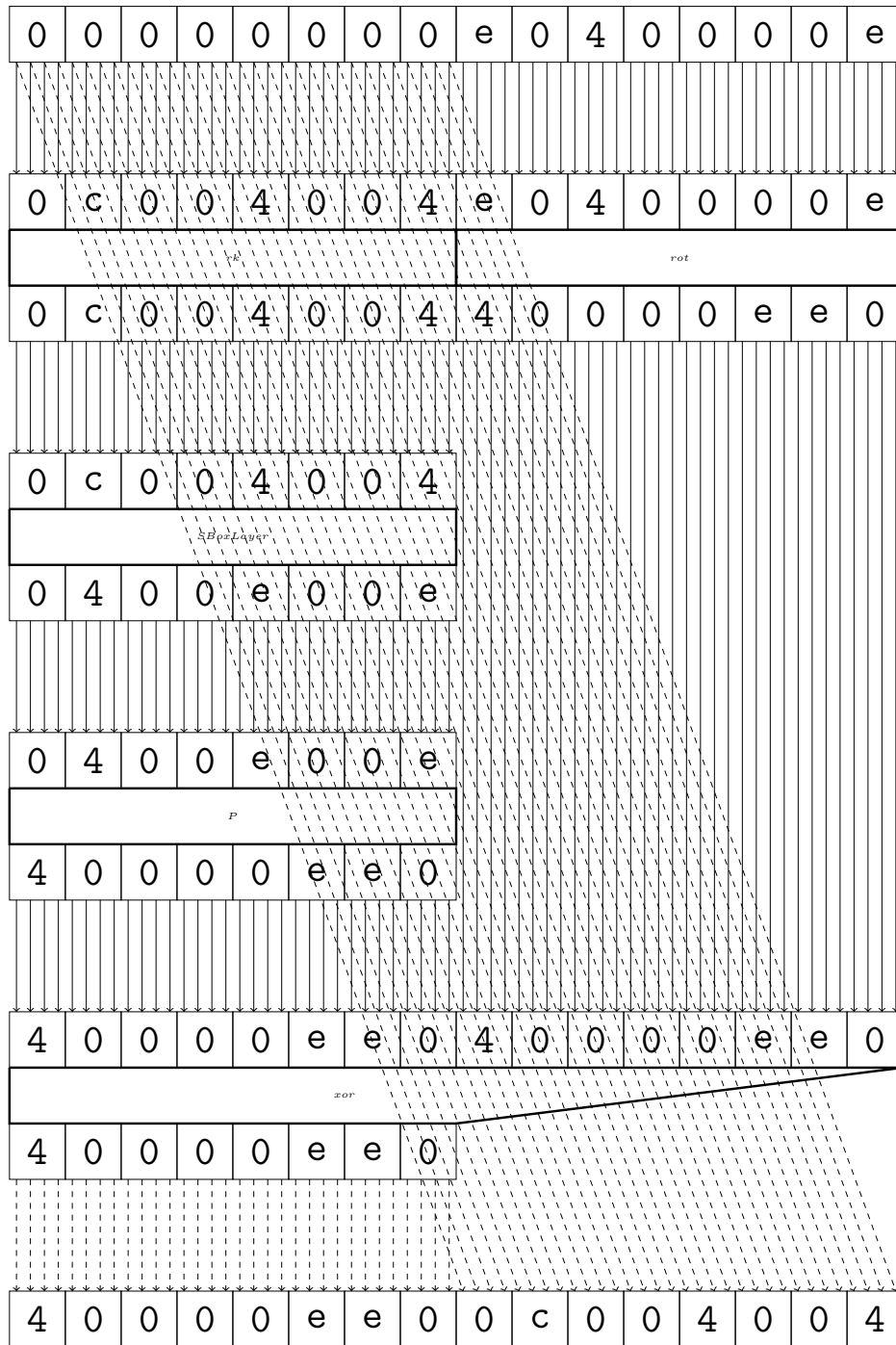
27 SBoxLayer

Maximal linear correlation: $2^{-0.0}$



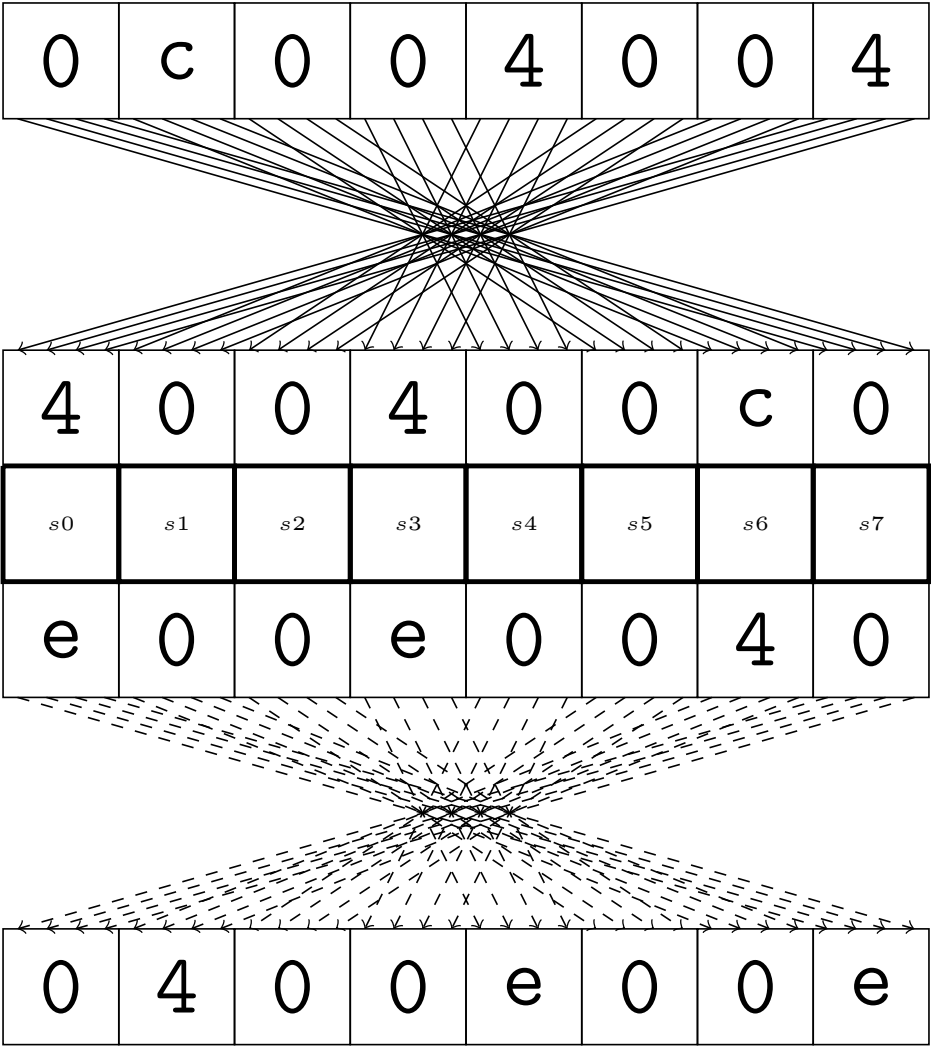
28 LBlock_round

Maximal linear correlation: $2^{-3.0}$



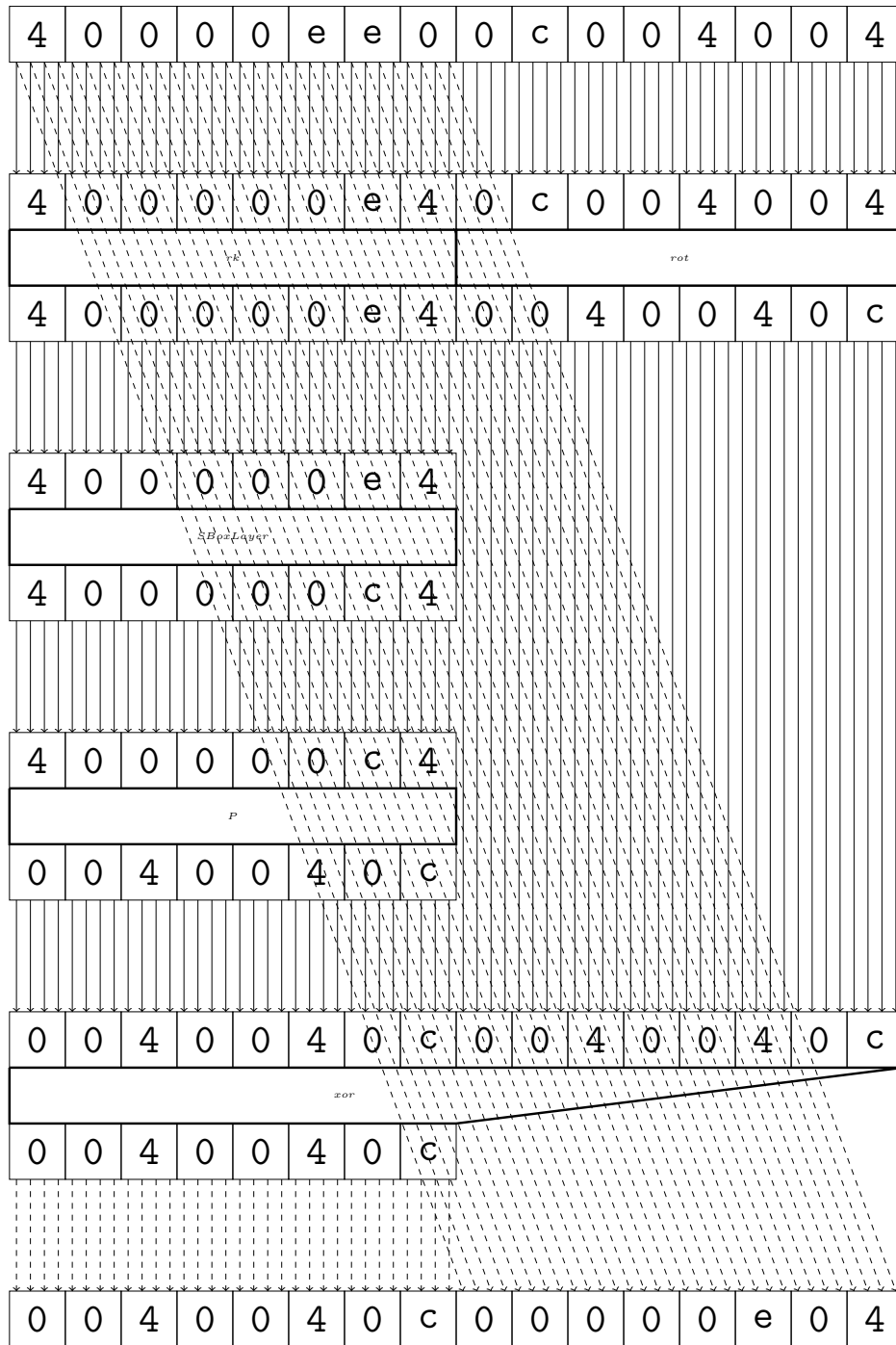
29 SBoxLayer

Maximal linear correlation: $2^{-3.0}$



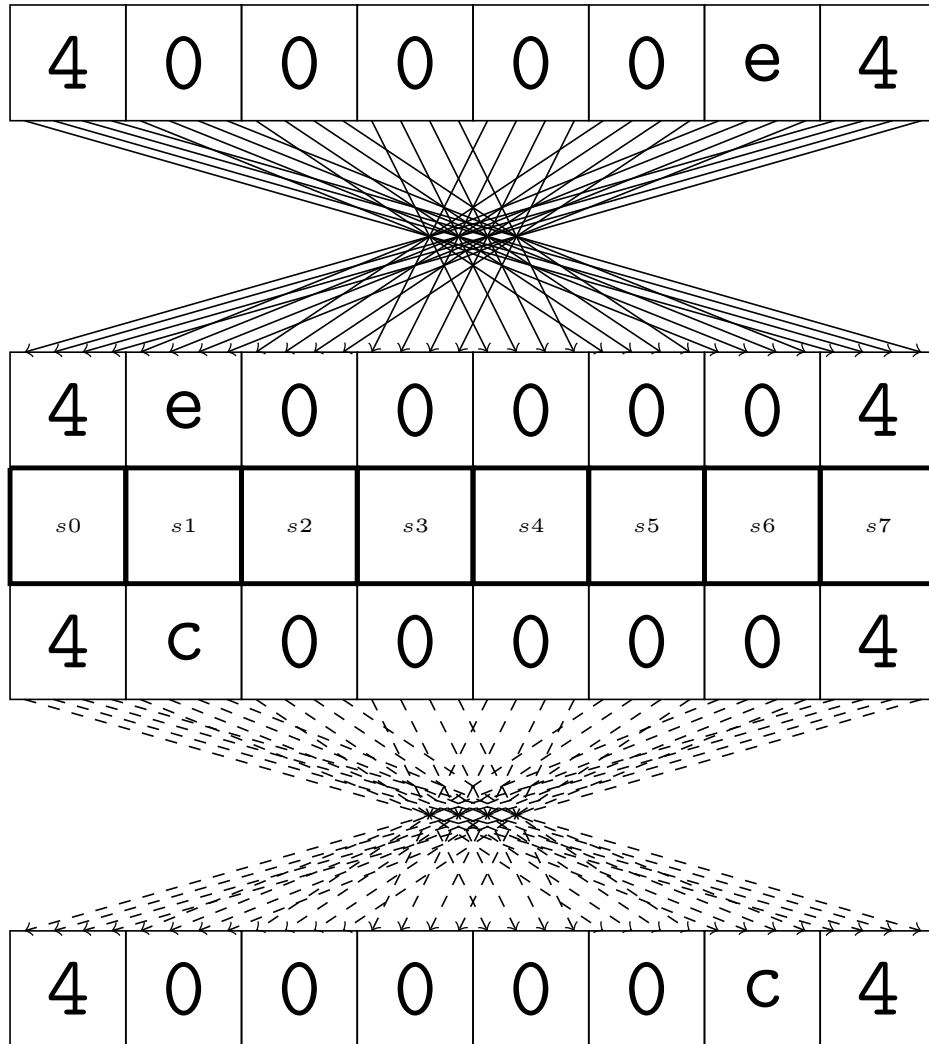
30 LBlock_round

Maximal linear correlation: $2^{-3.0}$



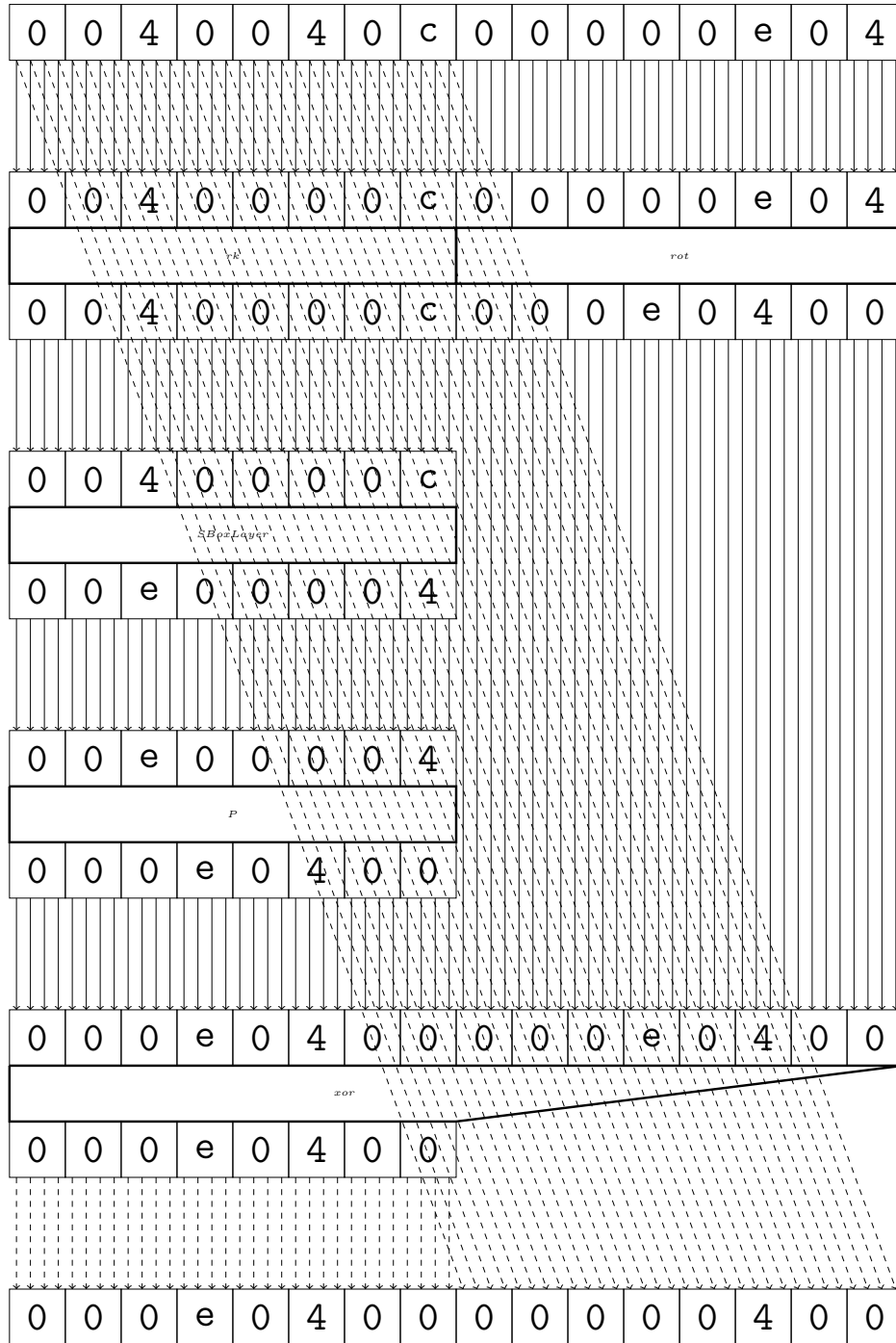
31 SBoxLayer

Maximal linear correlation: $2^{-3.0}$



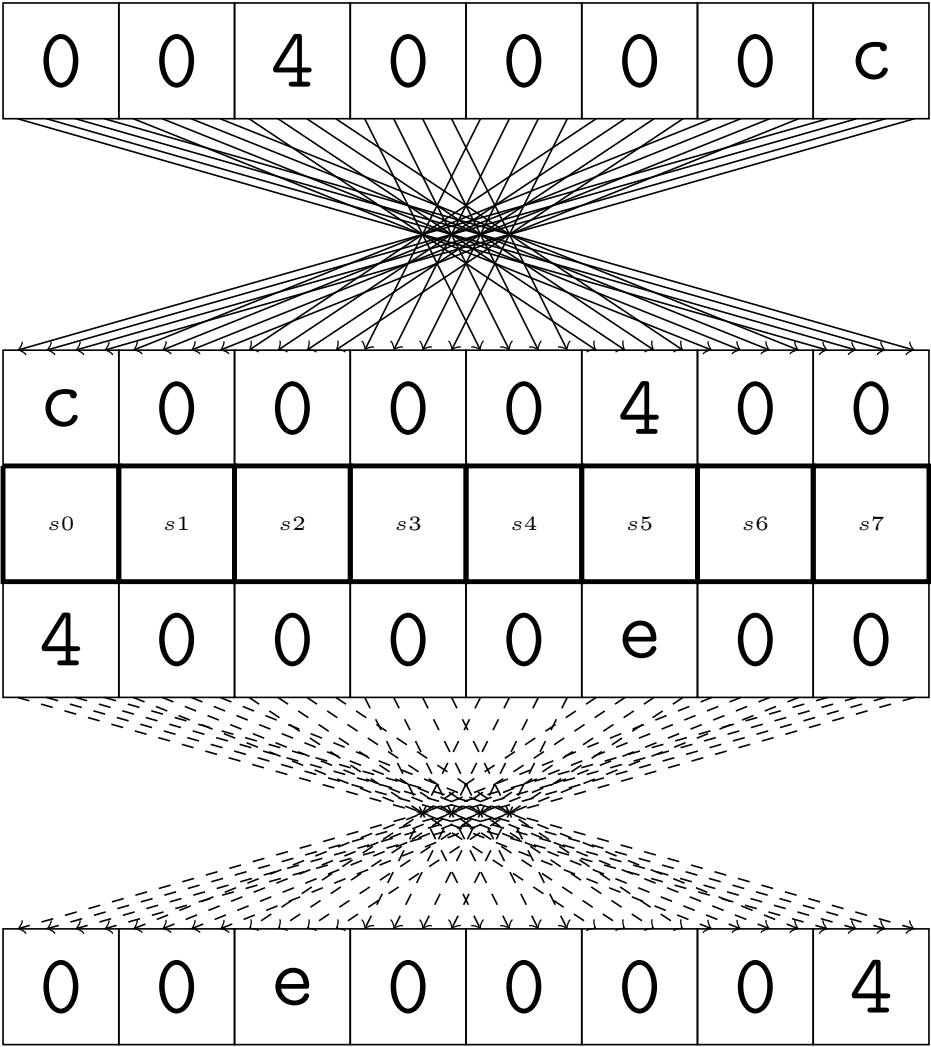
32 LBlock_round

Maximal linear correlation: $2^{-2.0}$

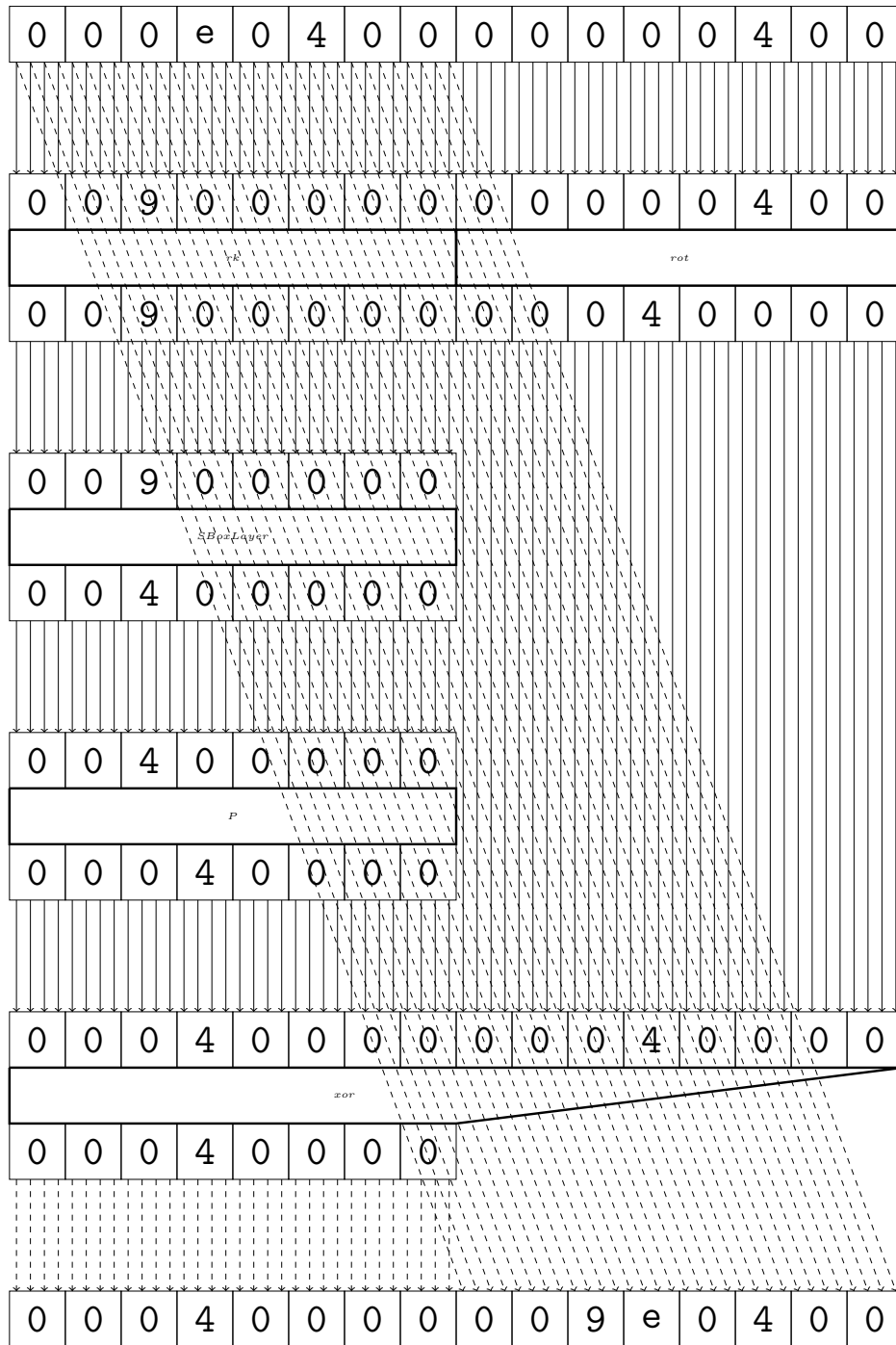


33 SBoxLayer

Maximal linear correlation: $2^{-2.0}$

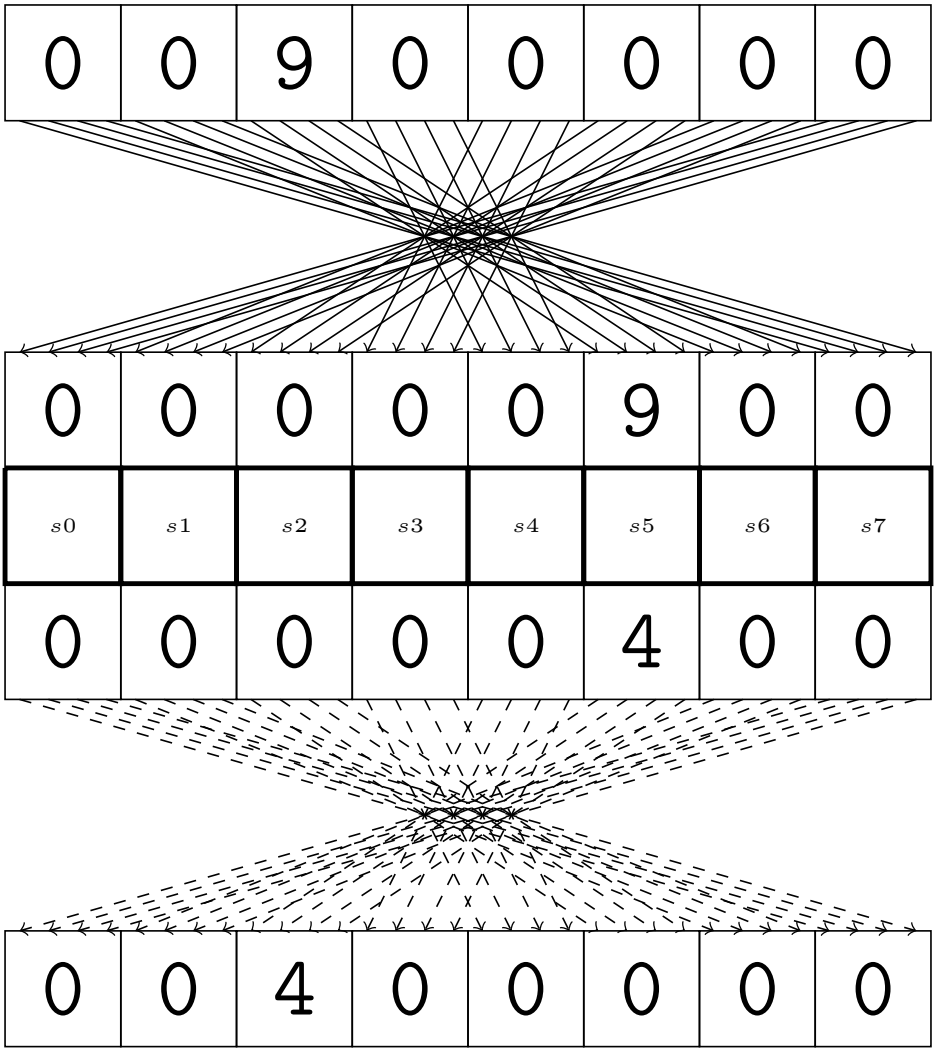


34 LBlock_round

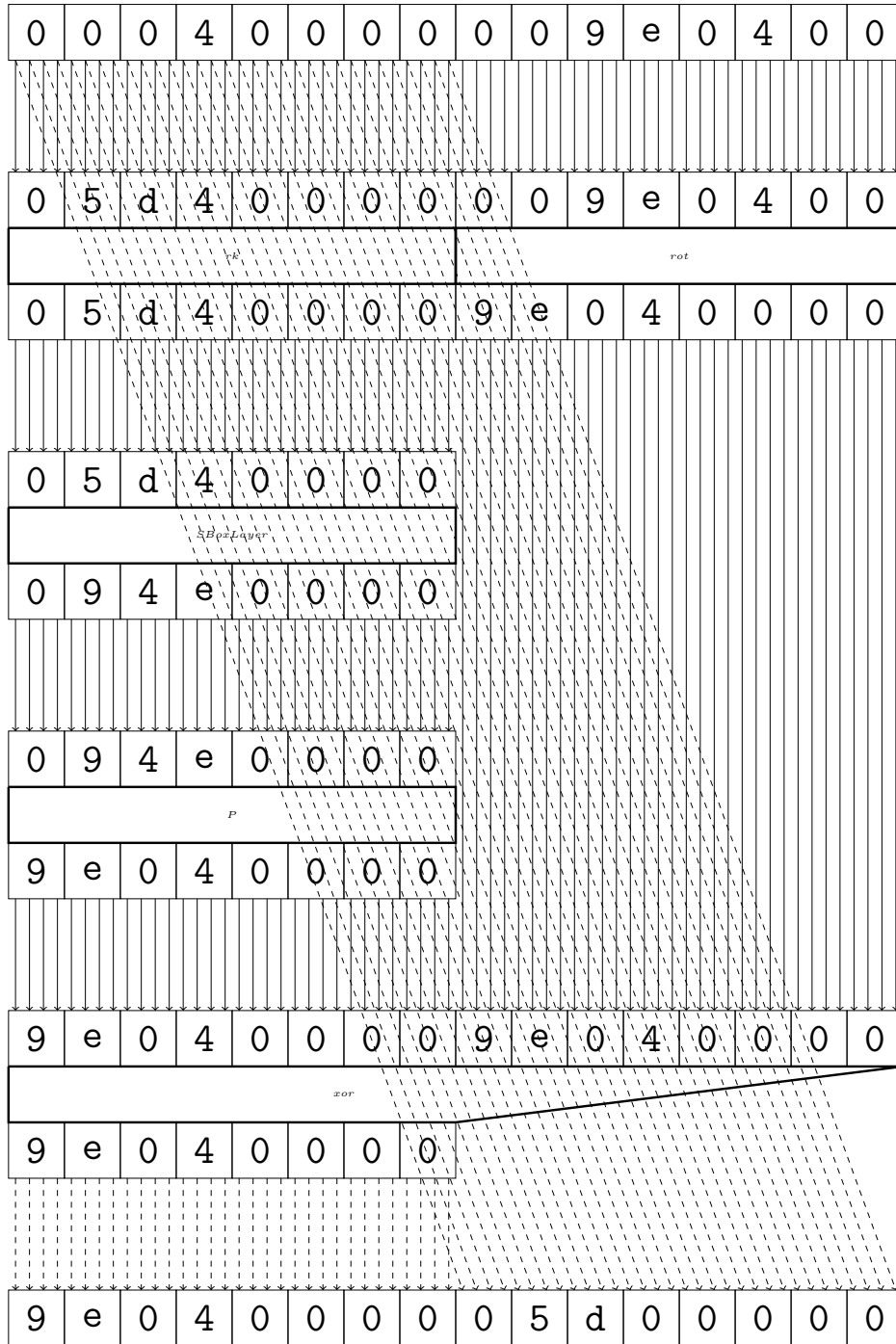
Maximal linear correlation: $2^{-1.0}$ 

35 SBoxLayer

Maximal linear correlation: $2^{-1.0}$

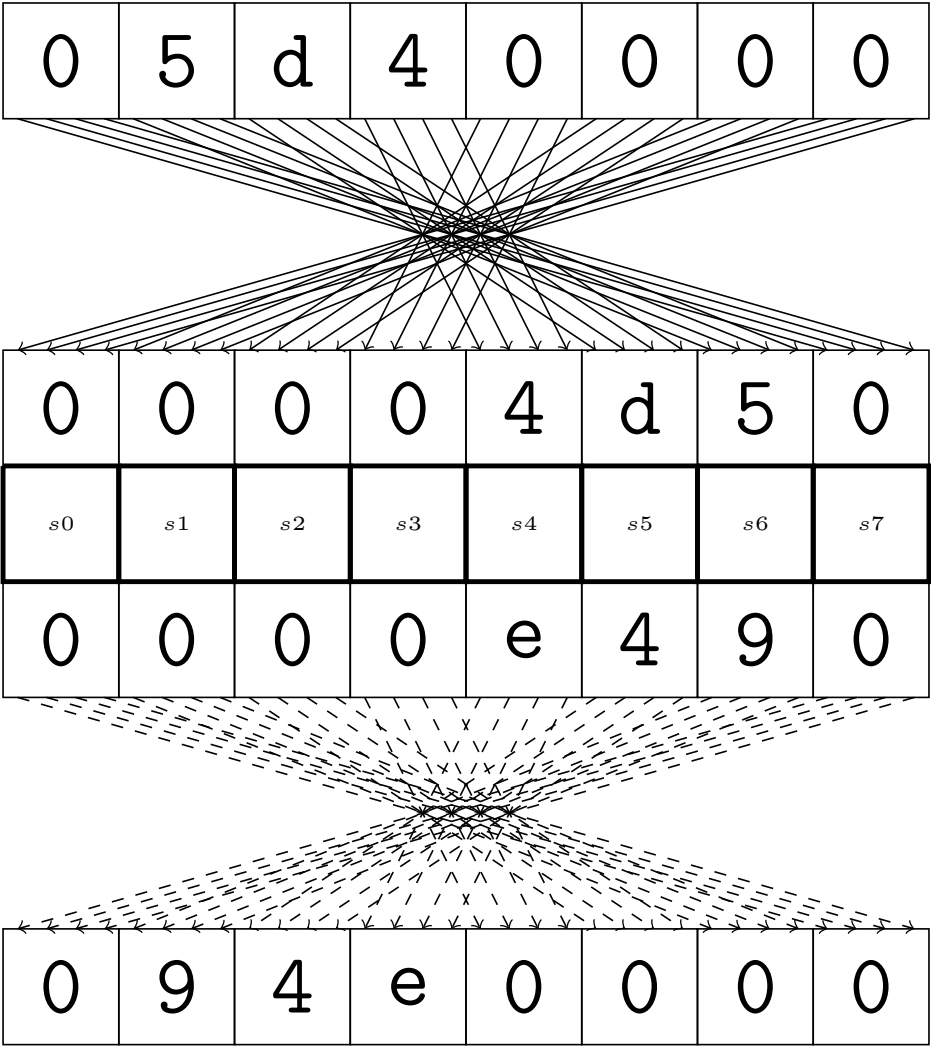


36 LBlock_round

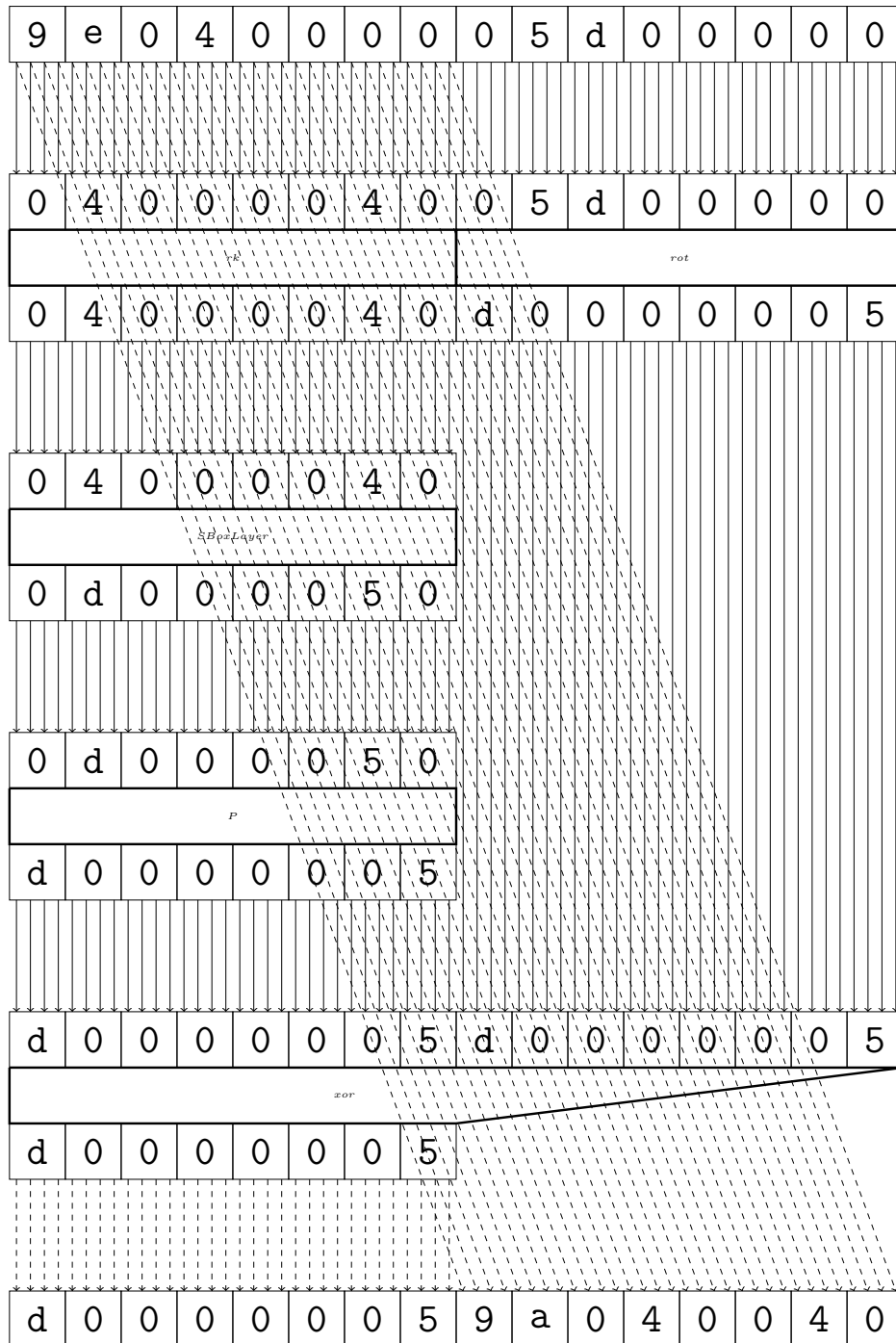
Maximal linear correlation: $2^{-3.0}$ 

37 SBoxLayer

Maximal linear correlation: $2^{-3.0}$



38 LBlock_round

Maximal linear correlation: $2^{-2.0}$ 

39 SBoxLayer

Maximal linear correlation: $2^{-2.0}$

